⑩ Ch 10 – Foreign Exchange Exposure and Risk Management

Asked in last 3 papers: every paper. Sep 2025 MCQ (broken-date swap points) + Q1(b) 4m (importer yen, forward vs unhedged) + Q6(b) 7m (money market hedge); Jan 2026 MCQ (early delivery: swap loss, interest on outlay) + Q2(a) 6m (import payment options) + Q3(b) 4m (cross rates, arbitrage); May 2026 MCQ (forward purchase: settle, close-out, rebook, early settle) + Q5(c) 2m (transaction vs translation).

1. Introduction — spot and forward market

→ **Exchange rate** = price of one nation's currency in terms of another (the **reference currency**). ₹ 100/$ → rupee is the reference currency; $ 0.01/₹ → dollar is.

→ **Spot rate**: delivery within **two business days** after the deal. **Forward rate**: rate fixed today for delivery on a future date (usually 30, 60, 90 or 180 days); payment and delivery at maturity.

→ (a) **Spot market**: India quotes rupees per one unit of foreign currency. (b) **Forward market**: buyer and seller agree today a specific amount, rate and future date; rate stays fixed whatever the spot does. The forward rate is **not** the future spot rate (which depends on demand and supply on that day).

→ Forex is the only market where **barter** happens — money delivered for money (selling £ forward = buying ₹ forward).

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Illus 1: one rupee buys **$ 0.0217**; **₹ 46.08** buys one dollar (reciprocals).

2. Role of SWIFT in Foreign Exchange

→ **SWIFT** (Society for Worldwide Interbank Financial Telecommunication): founded 1973, HQ La Hulpe (Belgium), non-profit; **purely a messaging system** — standardised communication, **not a settlement system**. Assigns **3-letter currency codes** (USD, GBP, CNY, CAD, AUD, HKD, INR, JPY, NZD, SGD, SEK, CHF, EUR). Connects institutions in 200+ countries.

→ **Why important: quick settlement** (messaging in seconds, plus validation and documentation); security and reliability → **lower operational risk**; standardisation → **efficiency and lower cost**; **full backup and recovery**; **catalyst** for collaboration among financial agencies.

3. National and International Payment Gateways

→ **Payment gateway**: virtual equivalent of physical cash transfer; an **encrypted channel** that authenticates and routes payment details from buyer's PC/mobile to banks for authorisation; completes the order with a reference number.

→ **Benefits:** 24x7x365; real-time card authorisation; rapid processing; multiple payment options; risk minimised by encryption; real-time reports; customer refund; merchants avoid complex software / huge data; CA-authenticated secure servers; cheap bulk data collection with card validity check; multiple host interfaces; simple administrative control; customers' support and merchants' trust.

→ **Challenges:** (a) customer may have no account with supporting banks; (b) limited banks in some gateways; (c) reliability, delivery, limited avenues, lack of trust in the provider.

→ **International gateways**: multi-currency, multi-language; customer pays in own currency → higher conversion, easy benchmarking of prices.

4. Nostro, Vostro and Loro Accounts

- Current accounts banks keep for quick transfer of funds in different currencies: "our", "your", "their".
- **Nostro** ("our account with you"): bank's FC account held abroad in that country's currency — Indian bank's CHF account in Switzerland.
- **Vostro**: foreign bank's account with a local bank — Citibank's account with SBI is Vostro for SBI. Nostro for one = Vostro for the other.
- **Loro**: using a third bank's Nostro account to settle — IDBI (no Nostro with Citi) asks HDFC (has account with Citi) to settle the import bill.

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TYK T2 (Nostro, Vostro, Loro) → this para.

4.1–4.2 Exchange Position and Cash Position

- **Exchange position**: total of purchase and sale commitments of the bank, **whether delivery has taken place or not** — entered on the contract date. Purchases > sales = **overbought**; sales > purchases = **oversold**.
- **Cash position**: balance in the **Nostro account** — credited for purchases, debited for sales, **only when actual delivery** takes place.
- Every cash-position item affects exchange position, **not vice versa**. Only ready deliveries (TT remitted) hit the Nostro; forward deals, bills purchased, cancellations hit only the exchange position.
- Cancellation of a forward purchase = a sale entry; cancellation of a draft sold = a purchase entry.

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Illus 2 (ABC Bank, £): purchases 35,000 + 12,500 + 40,000 + 15,000 (draft cancelled) = 1,02,500; sales 30,000 + 15,000 + 37,500 = 82,500 → **overbought £ 20,000**. Nostro 65,000 − 37,500 = **£ 27,500**. To keep Nostro £ 7,500: **sell £ 20,000 spot** (position → nil); to keep £ 7,500 overbought: **buy £ 7,500 forward**.

TYK P51 (CHF): purchases 1,60,000, sales 1,65,000 → **oversold 5,000**; Nostro 1,00,000 − 75,000 = 25,000 → **buy spot 5,000** (Nostro 30,000, position nil) and **buy forward 10,000**.

5.1–5.2 American / European terms; Direct and Indirect quote

- **American terms**: US$ per unit of foreign currency. **European terms**: foreign currency per US$ (most currencies are quoted this way), e.g. INR 44.92 per $.
- **Direct quote**: **home currency price of one unit of foreign currency** — $ 1 = ₹ 48.00 for an Indian. **Indirect quote**: foreign currency price of one unit of home currency — ₹ 1 = $ 0.0208.
- **Direct quote = 1 ÷ Indirect quote** (and vice versa). On reversing a two-way quote, bid and ask swap sides: 1/ask becomes the new bid (TYK P40: $/£ 1.5617–1.5673 → £/$ 0.63804–0.64033).

Note (app-written): SM 5.1 prints "U.S. dollar 0.2 per unit of Indian rupee" as the American quote beside INR 44.92/$; the reciprocal of 44.92 is about $ 0.0223.

5.3 Bid, Offer and Spread

- Two-way quote: **Bid** = rate at which the **dealer buys** the currency; **Offer/Ask** = rate at which he **sells**. ₹ 48.80–48.90/$: buys $ at 48.80, sells $ at 48.90. Offer is always higher — dealer buys at bid, sells at offer.
- A bid in one currency is simultaneously an offer in the other.
- **Spread % = (Offer − Bid) ÷ Bid × 100**.
- Dealer jargon: "**ONE MINE**" = I buy one million $ from you (at your ask); "**ONE YOURS**" = I sell one million $ to you (at your bid).
- **Customer side** (logic used in all TYK answers): customer **sells** FC → bank's **bid**; customer **buys** FC → bank's **ask**. Always the rate worse for the customer.

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TYK P2: Canara buys $ and credits ₹ to ABN-Amro's Vostro at interbank bid ₹ 51.3625 → $ 15,000,000 ÷ 51.3625 = **US$ 2,92,041.86** (or at $ 0.01947 → $ 2,92,050). **TYK P11 (i)** firm sells $ 2 months later at bid 47.00: 25,00,000 ÷ 47 = **US$ 53,191.489**; **(ii)** buys $ 2,00,000 spot at ask 46.25 = **₹ 92,50,000**.

S.4 Cross Rates

Asked: Jan 2026 Q3(b) 4m

- **Cross rate**: rate between a pair of currencies neither of which is the currency of the country where it is quoted (CAD/GBP quoted in India). In the Bloomberg table, rows are direct quotes for the first-column currency; columns are indirect quotes for the first-row currency.
- **Chain rule** (SM TYK P6, P3): **₹/$ x $/¥ = ₹/¥**; two-way: $(£/₹)_{ask} = (£/\$)_{ask} \times (\$/₹)_{ask}$. App-written generalisation: bid leg × bid leg gives the cross bid, ask leg × ask leg gives the cross ask; if one leg is quoted the other way, divide: ₹/HK$ = ₹ 42.85 / 7.5880 (TYK P7).
- Pick each leg from the side the bank actually deals on: when the bank buys the third currency, it buys $ at ask locally and sells $ for the third currency at the bid abroad.
- **Arbitrage test**: compute the cross rate; if a quoted rate differs, go round the markets (buy where cheap, sell where dear) and compare the end amount with the start.

SM item	Working	ICAI answer
TYK P3 (buy ₹ 25 m with £)	(£/₹) = (1/1.5260) × (1/161.3625) = £ 0.01068 ≈ 0.0107	**£ 2,67,500**
TYK P7 (HK$ 1 crore sold at ₹ 5.70)	Cover: buy $ at 42.85, sell $ for HK$ at 7.5880 → ₹ 42.85 ÷ 7.5880 = ₹ 5.6471	Profit **₹ 5,29,000**
TYK P8 (DKK 10 lakh sold at ₹ 6.5150 = ₹ 65,15,000)	London: 10,00,000 ÷ 11.4200 = £ 87,565.67 × 74.3200 = ₹ 65,07,881. New York: ÷ 7.5670 = $ 1,32,152.77 × 49.2625 = ₹ 65,10,176	Cover in **London**, profit **₹ 7,119** (NY ₹ 4,824; saving ₹ 2,295)
TYK P9 (SGD 25 lakh remitted late)	28 Jan: 45.90 × 1.7850 ÷ 3.1575 = 25.9482 + 0.125% (0.0324) = 25.9806. 4 Feb: 45.97 × 1.7775 ÷ 3.1380 = 26.0394 + 0.0325 = 26.0719	Loss to importer **₹ 2,28,250**
TYK P13 (3-market arbitrage, USD 1 crore)	$ → ₹ at 48.30 = ₹ 48,30,00,000 → £ at 77.52 = £ 62,30,650.155 → $ at 1.6231 = $ 1,01,12,968.26	Gain **USD 1,12,968.26**
TYK P46 (sold $ 10 lakh at € 1.9400, squared up at 1.9450)	€ 14,40,000 − € 14,45,000 = loss € 5,000; €/₹ = 31.4500 ÷ 1.9400 = ₹ 21.8403	Loss **₹ 1,09,201.50**
TYK P47 (i) (buy £ 1 m with CHF)	Buy $ from Bank A at CHF 1.4655; buy £ from Bank B at $ 1.7650 → 1.7650 × 1.4655 = CHF 2.5866	**CHF 25,86,600**

Note (app-written): TYK P3 — ICAI labels the rate $(£/₹)_{ask}$ but multiplies the reciprocals of $/£ 1.5260 and ₹/$ 61.3625; rounding 0.01068 up to 0.0107 gives £ 2,67,500, whereas 25,000,000 × 0.010679 ≈ £ 2,66,980. TYK P8 — the London rupee cost is printed "₹ 65,07,88"; the profit line shows ₹ 65,07,881.

S.5 Pips

- **PIP (Price Interest Point)** = smallest unit by which a quote can change (like tick size). USD/INR to customer ₹ 61.75 → pip 0.01; Indian interbank quotes to 4 decimals → 0.0001. EUR/USD 1.0213/1.0219 → spread 0.0006 = **6 pips**.

S.6 Forward exchange rate quotation

→ Market convention: forward quoted as **spot + forward margin (points)**, not as an outright rate. Spot 61.53/54, 6-month outright 61.93/98 → dealer quotes only **40/44** (61.93 − 61.53; 61.98 − 61.54). The "big figure" (61) is assumed known.

→ Forward margin is **not** the dealer's profit. Higher future price → **forward premium**; lower → **forward discount**.

→ Annualised: 40 paise per 6 months → 80 paise a year → 0.4 × 2 × 100 ÷ 61.53 = **1.30% p.a.** The forward market is really an **interest rate market** (compares the two currencies' interest rates).

Note (app-written): SM prints "61.98 − 62.54"; the 44 points come from 61.98 − 61.54.

S.7 Forward point determination

→ **Direct quote**: points **added** when the FC is at a **premium**, **deducted** at a **discount**. Indirect quote: exactly the opposite.

→ **How to tell without a sign**: spread must widen with time (uncertainty and liquidity). **Ascending points (5/8, 43/45)** → **premium** → **add**; **descending (25/20, 35/25)** → **discount** → **deduct**. GBP 1 = $ 1.6000–1.6010, points 5–8 → $ 1.6005–1.6018.

→ Points reflect the **interest rate differential**: the currency with the **lower interest rate is at a premium**; higher yield → cheaper forward. Aberration → arbitrage pushes back to equilibrium.

→ Points = pips of the forward; EUR/USD spot 1.1323, forward 1.1328 → 5 points (1 point = 0.0001).

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TYK P18: ₹ 48.5275 → 48.4575 = points (−) 0.0700 (discount); loan $ 20,00,000 + interest (SOFR 2% + 1% = 3% for 6 months) $ 30,000 = $ 20,30,000 × 48.4575 = **₹ 9,83,68,725**; market expects $ to depreciate — cover unless own view is a larger fall. **TYK P35:** spot 60.25–60.55, points 35/25 (descending → deduct) → 3-month **59.90/60.30**. **TYK P47 (ii):** GBP/CHF spot 1.4650 × 1.7645 = 2.5850 / 1.4655 × 1.7660 = 2.5881; GBP/USD 3m (25/20, discount) 1.7620/1.7640; USD/CHF 3m (5/10, premium) 1.4655/1.4665 → GBP/CHF 3m 2.5822/2.5869 → implied swap points **28/12 discount**. **TYK P16:** importer buys SGD 5 lakh on 30 Oct: USD/₹ 49.3800 + Spot/Oct 0.1300 = 49.5100; USD/SGD 1.7058 + 2nd-month 0.0096 = 1.7154; 49.5100 ÷ 1.7154 = 28.8621 + margin 0.05 = **₹ 28.9121**.

S.8 Broken period forward rate

Asked: Sep 2025 MCQ CS-IV 6m

→ Interbank forwards are for standard amounts and **standard month-end dates**; customer deals are odd amounts and odd dates (export bill realised 23 Feb). Points for the odd date are **interpolated** — assumed to grow **uniformly (pro rata)** over the period.

→ SM: premium for the odd date = **10*53/59** paise, "53 & 59 are broken & full periods", i.e. full-period points × broken period ÷ full period. Between two standard dates (TYK P5): (c) = (b) − (a) for the month between, (d) = (c)/2 for 15 days, (e) = (a) + (d).

→ Then: **merchant forward rate = spot ± broken points ± bank's margin** (margin deducted for exporter, added for importer).

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SM example: 1 Jan spot ₹ 62, 2-month (28 Feb) premium 10 paise → 62.10; margin 3 paise → importer 62.13, exporter 62.07. For 23 Feb: 10 × 53/59 = **8.98 paise** → 62.0898 ≈ 62.09 → exporter **62.06**, importer **62.12**.

TYK P5 (Sep 2025 MCQ type): 2m points 70/90, 3m 160/186 → 30 days 90/96 → 15 days 45/48 → 2m 15d = **115/138** (1 point = 0.0001) → rate for 20 June = 66.2525 + 0.0115 = **66.2640** / 67.5945 + 0.0138 = **67.6083**. Annual premium on average rate: bid 0.0115 ÷ 66.2583 × 12/2.5 × 100 = **0.0833%**; ask 0.0138 ÷ 67.6014 × 12/2.5 × 100 = **0.0980%**.

5.9 Merchant Rates

- → Nobody fixes the rate — market (demand and supply) decides. RBI does **not fix** but monitors **volatility** and intervenes by buying/selling USD. Wholesale **interbank rate** is the base for customer rates.
- → **Merchant rates** = rates to all customers (even an NRI remitting $) vs **interbank rates** between banks. FEDAI guidelines apply because of the customer-service element.
- → Till 1998 FEDAI prescribed margins; now banks set their own margin depending on their position. Only surviving rule: **FEDAI rule SA.8** — merchant settlements **rounded to the nearest whole rupee**: USD 1,234 × 61.32 = ₹ 75,668.88 → **₹ 75,669**.
- → History: gold standard ended 1971; rupee pegged to sterling, from 1975 to a basket; sterling schedule abolished 1984; FEDAI's margin / spread guidelines now redundant.
- → **Card rate**: merchant rates for import bill, export bill, inward/outward remittance, with standard spread, advised each morning to branches; used for walk-in and small deals. Regular / high-value customers get a better rate from the dealing room.
- → **Exchange margin** (in all SM illustrations): **deducted from a buying rate, added to a selling rate**; round as the question says (often to multiples of 0.0025).

6. Exchange Rate Forecasting

- → Needed for hedging, short-term financing and investment, capital budgeting, earnings assessment, long-term financing. Four groups:
- → **Technical**: historical data (time series); useful to speculators for day-to-day moves; rarely point/range estimates → limited use to MNCs.
- → **Fundamental**: relationships between economic variables and rates (subjective, regression, sensitivity). **Limits:** uncertain timing of impact; need to forecast factors with immediate impact; omission of non-quantifiable factors; sensitivity changes over time.
- → **Market-based**: current spot/forward rates (speculators make them reflect expected future rates).
- → **Mixed**: weighted average of several forecasts.

7. Exchange Rate Determination

- → Major factors: **inflation** (→ PPP) and **interest rates** (→ IRP). Others:
- → **Deficit/surplus on capital/current account**: current-account deficit → depreciation, surplus → appreciation; net capital inflow positive → appreciation, negative → depreciation.
- → **Trade barriers**: more import barriers/quotas → own currency appreciates in the long run.
- → **Central bank intervention** (buying/selling forex); **Government controls** (limits on FDI, FPI, repatriation).
- → **Expectations (Band Wagon effect)**: a dominant speculator goes short, others follow → currency falls.

8. Exchange Rate Theories — 8.1 Interest Rate Parity (IRP)

- → **IRP**: "size of the forward premium (or discount) should equal the **interest rate differential**". When IRP holds, **covered interest arbitrage** (FX risk covered by a forward) earns no more than a domestic investment.

Formula	Terms / logic	SM example figures
Covered IRP (SM): $(1 + r_D) = {}^{F}/_{S}\,(1 + r_F)$. App-written rearrangement: $F = S \times (1 + r_D) \div (1 + r_F)$. TYK P14 writes it as $F_1/E_0 = (1 + R_h)/(1 + R_f)$.	S, F = spot and forward in **direct** quote (home per foreign); r_D, r_F = domestic and foreign rates **for the period** (annual rate × months/12; TYK P14 uses 0.10/2, 0.04/2). Left = ₹ 1 invested at home; right = ₹ 1 converted at S, invested abroad, brought back at F. Equal → no arbitrage.	TYK P14: 55.50 × 1.05 ÷ 1.02 = **₹ 57.13**; $ at **premium** (India rate higher); premium (57.13 − 55.50)/55.50 × 12/6 × 100 = **5.87%**
Uncovered IRP (SM): $(1 + r_D) = {}^{S_1}/_{S}\,(1 + r_F)$	S_1 = **expected future spot** at which FC receipts are converted (no forward cover).	—
TYK P31 form: **$S_1 = S_0 \times (1 + \text{in A}) \div (1 + \text{in B})$**	Quote is units of currency A per unit of B; numerator = interest of the currency in which the price is stated.	TYK P31: £/$ 0.7570 × (1 + 0.075 × 3/12) ÷ (1 + 0.035 × 3/12) = 0.7570 × 1.01875/1.00875 = **£ 0.7645/$**. TYK P4: Can$ 2.50/£ × 1.075 ÷ 1.06 = **Can$ 2.535/£**

→ **Arbitrage rule** (SM 8.1): compare annualised forward premium/discount with the interest differential. If the forward discount on the high-interest currency is **less** than the differential → **borrow the low-rate currency, invest in the high-rate currency**, sell forward. If the discount is **larger** than the differential (excess premium on the low-rate currency) → invest in the **lower-interest** currency.

→ Steps: borrow → convert at spot → invest → convert maturity value at the forward → repay loan + interest → difference = gain.

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TYK P26: premium on $ (48.8190 − 48.0123)/48.0123 × 12/6 × 100 = 3.36% < differential 12% − 8% = 4% → borrow **$ 83,312** at 8% (repay $ 86,644.48), convert = ₹ 40,00,000, invest at 12% for 6 months = ₹ 42,40,000 ÷ 48.8190 = $ 86,851.43 → gain **US$ 206.95** (₹ 10,103).

TYK P27: DM premium (0.67 − 0.665)/0.665 × 12/3 × 100 = **3.01%** > differential 9% − 7% = 2% → borrow Can$ 1,000 at 9%, buy DM 1,503.76, invest at 7% → DM 1,530.08 × 0.67 = Can$ 1,025.15 − repay 1,022.50 = gain **Can$ 2.65**.

TYK P49: $ 5,00,000 for 3 months, cost 4%: London £ 3,24,886 + interest £ 4,061 = £ 3,28,947 less $ 5,05,000 ÷ 1.5430 = £ 3,27,285 → £ 1,662; New York $ 5,00,000 × (8% − 4%) × 3/12 = $ 5,000 ÷ 1.5475 = **£ 3,231**; Frankfurt cross 1.8260/1.5390 = € 1.1865 → € 5,93,250 + € 4,449 = € 5,97,699 × £ 0.5510 (1/1.8150) = £ 3,29,332 − 3,27,285 = £ 2,047 → invest in **New York**.

8.2 Purchasing Power Parity (PPP)

→ Focus: **inflation – exchange rate** relationship; rates reflect relative purchasing power of currencies.

→ **Absolute form ("Law of One Price")**: prices of similar products in two countries should be equal in a common currency; discrepancy shifts demand until prices converge. Alternative version allows **market imperfections** (transport costs, tariffs, quotas) via a **sectoral constant** — prices may differ, but their **rate of change** should be similar.

→ Problem: millions of products and constants; CPI baskets differ by consumption pattern (Finland vs Malaysia). **Relative form** overcomes market imperfections and consumption patterns.

→ PPP holds better in the **long run** and when disturbances are **purely monetary**.

Formula	Terms / logic	SM example figures
Absolute (equilibrium): $S = \alpha \times P_D \div P_F$	S = spot (₹/$); P_D, P_F = price levels in India / foreign; α = sectoral price and sectoral shares constant.	Bat ₹ 1,000 + transport ₹ 100 + US duty ₹ 200 → α = 1000/1300 = **0.7692**
Relative: $E(S_1) = S_0 \times (1 + I_d) \div (1 + I_f)$	S_0 = current spot (direct quote); I_d = home inflation; I_f = foreign inflation. Logic: a basket costing S_0 at home and 1 unit abroad grows by each country's inflation; rate = ratio of the two future prices. Higher-inflation currency depreciates. Can also be worked backwards.	₹ 50 × 1.12 = ₹ 56 vs $ 1.04 → **₹ 53.8462/$**. TYK P30: 46 × 1.08/1.04 → 47.77, 49.61, 51.52, **53.50** (years 1–4)
Real appreciation / depreciation = actual rate − PPP rate	Any difference from the PPP rate is explained by factors other than inflation.	45 × 1.10/1.02 = 48.53; actual 50.00 → USD real appreciation **₹ 1.47** = 1.47/45 = **3.27%**

8.3 International Fisher Effect (IFE)

- → Uses **interest rate** (not inflation) differentials to explain exchange-rate changes; closely related to PPP since interest rates are highly correlated with inflation.
- → Nominal risk-free rate = **real return + anticipated inflation**. If all investors want the same real return, interest differentials reflect expected inflation differentials. **High-interest currencies will depreciate**. Nominal rate also includes default risk.
- → **Equation:** $r_D - P_D = r_F - \Delta P_F$, or $\mathbf{P_D - P_F = \Delta S = r_D - r_F}$. Terms: r = nominal interest, P = inflation, ΔS = % change in spot. Logic: with free capital flows, investment flows until **real returns equalise**.
- → If IFE holds, borrowing in one country and investing in another gives **no positive return on average**. It does not consistently hold because PPP does not (other factors affect rates).

8.4 Comparison of PPP, IRP and IFE

Theory	Key variable	Basis	Summary
IRP	Forward premium/discount	Interest rate differential	Why forward differs from spot, at a **point of time**; covered interest arbitrage gives no more than a domestic return.
PPP	% change in spot rate	Inflation differential	Spot changes **over time** with inflation differential; consumers' purchasing power at home = when importing.
IFE	% change in spot rate	Interest rate differential	Spot changes over time with interest differential; uncovered foreign money-market return on average no higher than domestic.

9. Foreign Exchange Market

- → Market where individuals, firms and banks buy and sell foreign currencies; purpose: **transfer of purchasing power** from one currency to another (trade invoiced in one agreed currency, possibly a third one like $).
- → Demand for FC: tourists going abroad, imports, investing abroad. Supply: foreign tourists, exports, foreign investment received. Commercial banks act as **clearing houses**.
- → **Four levels:** (1) tourists, importers, exporters, investors; (2) commercial banks; (3) foreign exchange brokers (banks even out flows among themselves); (4) central bank — lender/buyer of last resort (draws down or adds to reserves).
- → **Participants:** non-bank entities (MNCs, individuals); banks; **speculators** (profit from rate fluctuations); **arbitrageurs** (profit from price differences in two markets simultaneously); governments via central banks (stabilise rates).

10. Foreign Exchange Exposure

Asked: May 2026 QS(c) 2m

→ **Exposure**: "a contracted, projected or contingent cash flow whose magnitude is not certain at the moment; depends on FX rates and interest rates". FX exposure = parts of business affected if exchange rates change.

→ Examples: traveller (destination currency appreciates); **exporter** invoicing in FC (FC falls → lower home revenue); **importer** (FC appreciates → higher cost); fund managers owning foreign assets (currency falls); trade in local currency where rates move the negotiated price.

→ Timing view: **translation** and **transaction** exposure look at the moment of the rate change (accounting / settled obligations); **operating (economic)** exposure looks at expected future cash flows after an unexpected change.

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TYK T1 ("operations in foreign exchange are exposed to a number of risks") → this para 10 (three exposures). **TYK P1:** bond return (5,250 − 5,000 + 350)/5,000 = 12% for a US investor; non-US investor with $ up 2%: 1.12 × 1.02 − 1 = **14.24%**.

10.1 Transaction Exposure

→ Effect of a rate change on **outstanding obligations** entered before the change but **settled after** it — cash flows from **existing contractual obligations**. Arises whenever there are FC receivables/payables.

→ Exporter with $ 1,00,000 receivable in 6 months: $ depreciates → cash loss; appreciates → gain. Budgeted ₹ 62/$, falls to ₹ 57 → commercial profit wiped off. Most common exposure; runs from budgeting until extinguished by sale/purchase of FC.

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TYK P48 (A) (Omega, 2,400 units): profit at current rates 2400 [€ 500 × 51.50 − (S$ 800 × 27.25 + 1,000 + 1,500)] = ₹ 34,80,000; after change (52 and 27.75) ₹ 31,20,000 → loss due to transaction exposure **₹ 3,60,000**. **TYK P10 (MNP, net exposure):** net inflow × (forward − spot): $ 20 m × 0.81 = 16.20; FFr 12 m × 0.67 = 8.04; £ 10 m × 0.41 = 4.10; yen −10 m × −0.80 = 8.00 (₹ million); yen net payable with yen at discount, others net receivable at premium → positions offsetting.

Note (app-written): TYK P10 asks for "net exposure in terms of rupees" but ICAI multiplies the net inflow by the forward-minus-spot difference ("spread"), which is the gain from the forward differential rather than the rupee value of the exposure. Keep ICAI's figures.

10.2 Translation Exposure

→ Gains/losses from **translating** foreign currency assets and liabilities into the parent's currency for **consolidation**; also called **accounting exposure** — accounting-derived changes in owner's equity.

→ SM example: $ 1 m loan for imported capital goods at ₹ 55 → asset and loan ₹ 550 lacs; at year end ₹ 58 → translation loss **₹ 30 lacs**, capitalised → asset ₹ 580 lacs → higher depreciation, lower profit.

→ Both transaction and translation exposures hit the bottom line (gain if movement favourable). **Translation exposure becomes transaction exposure** at some stage — the loan must be repaid by buying $.

10.3 Economic Exposure

→ Extent to which the **economic value of the company** can decline due to rate changes — overall impact on firm value; rate changes alter input costs and output prices → **competitive position**.

Cash inflow item (local currency)	Local revaluation	Local devaluation
Local sales relative to foreign competition	Decrease	Increase
Exports in local currency / in foreign currency	Decrease	Increase
Interest from foreign investments	Decrease	Increase
Cash outflow item	**Revaluation**	**Devaluation**
Imports denoted in local currency	Same	Same
Imports denoted in foreign currency	Decrease	Increase
Interest on foreign debt	Decrease	Increase

→ **Why FX risk management matters:** (i) **protection against volatility** (lock rates in advance); (ii) **cost reduction**; (iii) **competitive advantage** (better prices and payment terms); (iv) **improved cash flow** (visibility, predictability); (v) **compliance with regulations** (avoid fines).

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TYK P48 (B) (operating exposure, price ₹ 25,000): profit at new rate (S$ 27.15) 2400 × 780 = ₹ 18,72,000; after 6 months (27.75) 2400 × 300 = ₹ 7,20,000 → decline **₹ 11,52,000** (transaction). € price 25,000/51.50 = € 485.44 → 25,000/51.75 = € 483.09, fall € 2.35 (0.48%) × elasticity 1.5 = 0.72% → 2,417 units → profit ₹ 7,25,100 → decrease due to operating exposure **₹ 11,46,900** (if fixed cost unchanged: ₹ 7,42,100 → ₹ 11,29,900).

II. Hedging Currency Risk — (A) Internal techniques

→ **Natural hedging** by exporters: short-term FC loans — **PCFC** (pre-shipment credit in FC) / **FCNR(B)** loans: (1) cheaper than domestic-currency rates; (2) export collections settle the PCFC loan → transaction risk hedged.

→ Internal techniques involve **no explicit transaction cost**; offset exposure fully or partly:

→ (i) **Invoicing in domestic currency**: avoids exposure but currency is part of the marketing package; buyer may prefer its own; markets like oil/aluminium follow competitors' currency; in a buyer's market sellers invoice in the buyer's currency. If invoicing in FC, choose a **major currency with an active (deep) forward market** at least as long as the credit period; avoid limited-convertibility, chronically weak currencies.

→ (ii) **Leading and lagging**: leading = pay before due date; lagging = delay after due date — to gain from expected devaluation/revaluation. Payer's currency expected to devalue → **lead**; expected to revalue → **lag**.

→ (iii) **Netting**: group companies settle inter-affiliate debts for the **net amount**. **Bilateral netting** between pairs; flows reduced by the lower of purchases/sales. Main problem: settlement currency. Reduces banking costs, spreads and charges; increases central control.

→ (iv) **Matching**: matches FC inflows with FC outflows by amount and timing; applies to **intra-group and third parties** (netting = within group only). Needs two-way flows in the same currency (**natural matching**); **parallel matching** = different currencies expected to move together. Both need enabling exchange control rules — an MNC subsidiary in India cannot net receivables and payables.

→ (v) **Price variation**: raise selling prices against adverse rate moves; in some countries the only legal tactic. **Transfer price variation** in inter-company trade — but tax, customs and exchange control authorities require **arm's length price**.

→ (vi) **Asset and liability management**: for balance sheet, income or cash-flow exposure (cash-flow focus makes sense; pure translation focus misplaced). **Aggressive**: increase exposed inflows in strong currencies / outflows in weak ones. **Defensive**: match inflows and outflows by currency regardless of strength.

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TYK P43 (lagging, NP and Co.): (A) cover both: payable $ 7,00,000 × 48.45 = ₹ 3,39,15,000 − receivable $ 4,50,000 × 48.90 = ₹ 2,20,05,000 − interest 12% for 1 month ₹ 2,20,050 = **₹ 1,16,89,950**. (B) lag receivable 1 month, cover net $ 2,50,000 × 48.45 = ₹ 1,21,12,500, cancellation profit (48.90 − 48.30) × 4,50,000 = ₹ 2,70,000 → **₹ 1,18,42,500** → option (A) preferred. **Matching — TYK P38:** £ 1,38,000 receipt matched with £ 4,80,000 payment (both 3 months) → hedge only net £ 3,42,000.

Note (app-written): TYK P43 — ICAI reads "25/30" and "40/45 points" as outright rates 48.25/48.30 and 48.40/48.45 (its Note 1), and labels the net as "(A) + (B)", but the ₹ 2,70,000 profit is deducted: 1,21,12,500 − 2,70,000 = 1,18,42,500.

II. (B) External techniques — (i) Money Market Hedge (MMH)

Asked: Sep 2025 Q6(b) 7m

→ Fixes today the home-currency value of a future FC flow by **borrowing and investing in the money markets** at today's spot (SM: owner expecting $ 1 m in six months locks the INR value now).

→ **Advantages:** (a) fixes future rate — no downside risk; (b) **flexible amount**; (c) usable where **forwards are not available**. **Disadvantages:** (a) **more complicated** than a forward; (b) no benefit from favourable movements.

Step (SM TYK P29 wording)	FC receivable (exporter)	FC payable (importer)
1 Identify / Create	FC is an asset → create an FC **liability**	FC is a liability → create an FC **asset**
2 Borrow / Deposit FC	**Borrow FC** = receivable ÷ (1 + FC borrowing rate for period)	**Deposit FC** = payable ÷ (1 + FC deposit rate for period)
3 Convert at spot	Sell FC at the rate worse for you (bid in direct quote)	Buy FC with home currency (ask in direct quote), borrowed at home
4 Invest / Borrow home	Invest home currency at home **deposit** rate	Home loan repaid at home **borrowing** rate
5 Settle	Customer's FC repays the FC loan; keep home deposit maturity	FC deposit matures = payable; pay home loan + interest
6 Compare	With forward receipt (higher is better)	With forward cost (lower is better)

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TYK P28 (Rohit, £ 5,00,000 receivable): borrow £ 5,00,000 ÷ 1.0125 = £ 4,93,827 × ₹ 56 = ₹ 2,76,54,312; invest at 12% for 3 months = **₹ 2,84,83,941**; gain ₹ 4,83,941 over ₹ 56 × 5,00,000.

TYK P29 (UK exporter, $ 3,50,000): borrow at 9% (2.25%/quarter) $ 3,42,298.29; convert at ask 1.5905 (indirect quote) = £ 2,15,214.27; invest at 5% → **£ 2,17,904.45**; forward 3,50,000 ÷ 1.6140 = £ 2,16,852.54 → MMH better by **£ 1,051.91**.

TYK P38 (Columbus, payable £ 3,42,000): forward 3,42,000 ÷ 0.9520 = $ 3,59,244; MMH deposit £ 3,42,000 ÷ 1.025 = £ 3,33,658 ÷ 0.9830 = $ 3,39,429 loan × 1.0325 = **$ 3,50,460** → MMH cheaper. **Receipt € 5,90,000 (4 months):** forward × 1.9510 = **$ 11,51,090**; MMH borrow € ÷ 1.0533 = € 5,60,144 × 1.8890 = $ 10,58,113 × 1.0383 = $ 10,98,639 → forward better.

TYK P39 (b): £ 3,00,000 ÷ 1.045 = £ 2,87,081 × $ 2 = $ 5,74,162 × 1.055 = **$ 6,05,741**. **TYK P40 (ii):** $ 3,64,897 ÷ 1.0225 = $ 3,56,867 × 0.64033 = £ 2,28,512 + 3.5% interest £ 7,998 = **£ 2,36,510**.

II. (B)(ii) Derivative Instruments

→ **Derivative**: bilateral contract whose value **derives from** an underlying asset, reference rate or index (interest rates, FX, commodities, equities). Includes OTC deals, exchange-traded futures and options on futures, call warrants. Transaction risk hedged by **forwards, futures, options, swaps**.

12. Forward Contract

Asked: Sep 2025 Q1(b) 4m **Asked: Jan 2026 Q2(a) 6m**

→ Simplest derivative: obliges one party to buy and the other to sell a specified quantity at a **specific price on a specified future date**. FX forwards, FRAs, commodities. Importers can cover all liabilities by forwards.

→ **Hedge choice** (all SM answers): exporter **sells FC forward** (bank's bid); importer **buys FC forward** (bank's ask). Compare: forward outcome vs unhedged at expected spot.

→ **Expected spot** with probabilities = Σ (rate × probability) (TYK PIS, P39).

→ **Pay now vs credit**: pay now = spot cost + borrowing interest for the period; pay later = (FC + supplier's interest) × forward rate (+ bank loan interest for any gap). Choose the lower outflow.

SM item	Working	ICAI answer
TYK P17 (importer, 108 lakh yen) — Sep 2025 Q1(b) type	Spot 108/30 = 3.6 yen/₹; −10% → 3.24 → cost 33.33 lakh; forward 3.3 → 32.73 lakh	Loss unhedged **₹ 3.33 lakh**; with forward **₹ 2.73 lakh** → take forward
TYK P6 (JKL, export ¥ 1 crore)	Cross ₹/¥: 62.22/102.34 = 0.6080; expected 65/124 = 0.5242; forward 66.50/110.35 = 0.6026; actual 66.25/110.85 = 0.5977	Unhedged loss **₹ 8,38,000**; forward loss ₹ 54,000; actual unhedged loss ₹ 1,03,000 → forward justified
TYK P4 (ABC UK, Can$ 5 lakh)	Forward Can$ 2.535 → £ 1,97,239. Spot 2.55 → £ 1,96,078; 2.40 → £ 2,08,333; 2.50 → £ 2,00,000	Gain £ 1,161; loss £ 11,094; loss £ 2,761
TYK P11 (iii) ($ 69,000)	Now 69,000 × 46 = 31,74,000 + 10% for 2 months 52,900 = 32,26,900; forward 69,000 × 47 = 32,43,000	Encash **after 2 months**
TYK P15 (Multinational)	EV = 0.24 + 0.34 + 0.45 + 0.38 + 0.40 = $ 1.81 vs forward $ 1.80	**Retain** receivable (don't sell forward)
TYK P12 (Z Ltd, USD 2 m) — Jan 2026 Q2(a) type	(i) 2 m × 57.10 = ₹ 11,42,00,000 + loan 10% 30 days ₹ 9,51,667; (ii) (2 m + 8% 30 days $ 13,333) × 57.50	(i) **₹ 11,51,51,667** vs (ii) ₹ 11,57,66,648 → pay in 60 days
TYK P22 (DEF, US$ 1 crore)	(i) 63.15 cr + 9.5% 30 days 0.50 cr; (ii) 1.00646 cr × 63.45	(i) **₹ 63.65 cr** vs (ii) ₹ 63.86 cr → (i)
TYK P21 ($ 1,30,000)	Now: 1,30,000 × 48.36 = 62,86,800 + OD 15% 3 months 2,35,755 = 65,22,555. Later: 1,31,625 × 48.83	Later **₹ 64,27,249** → pay after 3 months
TYK P35 (Gibralter, $ 1,00,000)	(i) $ 1,02,500 × 60.30 = 61,80,750; (ii) 1,00,000 × 60.55 = 60,55,000 + 14% 3 months 2,11,925	(i) **₹ 61,80,750** vs ₹ 62,66,925 → (i)
TYK P42 (Sun, 3,400 lakh yen)	Loan 18% quarterly: 1,000 × 1.045² = 1,092.03. LC: commission 10.00 + interest 0.90 = 10.90; 3,433.53 lakh yen ÷ 345 × 100 = 995.23	**₹ 1,006.13 lakh**; cheaper by ₹ 85.90 lakh → accept
TYK P19 (Excel, $ 1 lakh)	(45.20 − 45.60)/45.60 × 365/60 × 100; loss (45.20 − 45.50) × 1,00,000	Discount **5.33%**; loss **₹ 30,000**
TYK P23 ($ 60,000, forward 64, 2% upfront)	Premium 60,000 × 64 × 2% = 76,800 + 12% for 6 months 4,608 = hedging cost 81,408	At 68: gain 1,58,592; 62: loss 2,01,408; 70: gain 2,78,592; 65: loss 21,408
TYK P24 (AKC, indirect quotes)	Hedged at 3m forward ask (2.427, 0.0216, 0.0178) = ₹ 1,33,38,719, contribution 26,08,719; unhedged at 3m spot ask = ₹ 1,32,75,578, contribution 25,45,578	C/S **19.56%** vs 19.17% → hedge

12.1 Forward Rate — Premium and Discount

→ Forwards: **customised**, not standardised (size, grade, place, date, credit period negotiated). Value change ≈ value change in the underlying; payoff profile differs from options. Buyer gains if price at maturity > contract price; buyer's gain = seller's loss.

→ **Premium**: currency costlier forward → in direct quote **added** to both buying and selling rates. **Discount**: cheaper forward → **deducted**.

→ **Annualised premium/discount = [(Forward Rate − Spot rate) / (Spot rate)] x (12/6)** (SM, 6-month example; × 12/months of the forward, × 100 for %; TYK P19 uses × 365/60 for days). Sign + = premium, − = discount. Work it for each currency in its own direct form.

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SM example: spot ₹ 55, 6-month ₹ 60 → $ premium (60 − 55)/55 × 12/6 = **18.18%**; rupee: (0.01667 − 0.01818)/0.01818 × 12/6 = **−16.61%** (discount). Premium ≠ discount.

Note (app-written): the SM computes the rupee discount as −16.61% and then says "16.67%"; the gap is rounding (exactly (55/60 − 1) × 2 = −16.67%).

12.2 Fate of Forward Contracts

Asked: Jan 2026 MCQ CS-II 6m Asked: May 2026 MCQ CS-V 6m

→ Three fates, each on due date / early / late: **(A) Delivery, (B) Cancellation, (C) Extension.**

→ **Core rule — cancel by the reverse deal**: bank's **sale** contract (importer) is cancelled at the bank's **buying** rate; bank's **purchase** contract (exporter) at the bank's **selling** rate. Margin deducted from buying, added to selling. Difference vs contract rate is charged to / paid to the customer (+ flat charges).

Fate	Rate applied (SM)	SM figures
Delivery on due date	Contract rate, whatever the spot.	Illus 3: $ 1,00,000 at 65.5000 → debit **₹ 65,50,000**
Early delivery	Contract rate + charges: (a) **swap difference** (bank undoes its cover: sells/buys spot now, reverses at forward for original date); (b) **interest on outlay of funds** at not less than **PLR** from early date to due date (if inflow, bank may pass on term-deposit interest).	Illus 4 below
Late delivery	Current (spot) rate on delivery date, after applying **automatic cancellation**.	—
Cancellation on due date	Reverse at **spot** (TT) rate ± margin; difference + flat charges.	Illus 5 below
Early cancellation	Reverse at the **forward rate for the original due date** (opposite side).	Illus 6; TYK P44, P45
Late cancellation	Automatic cancellation provisions.	—
Extension on due date	Cancel at spot (as cancellation on due date) + **rebook** at forward for the new period.	Illus 7; TYK P32
Extension before due date	Cancel at forward for original date + rebook at current forward for new date.	Illus 8
Late extension	Current rates after automatic cancellation.	Illus 9

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Illus 4 (Mr. X exporter, $ 1,00,000 at 65.40, delivered 28/30 Nov instead of 31 Dec; spot 65.22/65.27, 1-month 10/15, PLR 18%): (a) **swap loss**: bank sells spot 65.22, buys forward 65.27 + 0.15 = 65.42 → 0.20 × 1,00,000 = **₹ 20,000**; (b) **outlay**: bank pays customer 65.40, sells spot 65.22 → 0.18 × 1,00,000 × 31/365 × 18% = **₹ 275**; charges ₹ 20,275; net inflow 65,40,000 − 20,275 = **₹ 65,19,725**.

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Illus 5 (import $ 2,50,000 at 65.3450 cancelled on due date): spot buying 65.2900 − 0.10% (0.0653) = 65.2247 ≈ **65.2250**; 65.3450 − 65.2250 = 0.12 × 2,50,000 = ₹ 30,000 + flat ₹ 100 = **₹ 30,100**.

Illus 6 (purchase AUD 1,00,000 at 47.2500, cancelled after 2 of 3 months): at 1-month forward sale 47.5200 → 0.27 → **₹ 27,000**. **TYK P44:** CHF 10,000 at 27.25, cancelled at 1-month sale 27.52 → **₹ 2,700**. **TYK P45:** USD/₹ 49.4455 + April 0.4200 = 49.8655 + 0.10% = 49.91537; ÷ 1-month CHF 1.5150 = 32.9474 ≈ 32.9475 − 32.4000 = 0.5475 → **₹ 54,750**.

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Illus 7 (export $ 1,00,000 at 62.5200, extended on due date by 1 month): cancel at spot selling 62.7200 + 0.10% = 62.78272 ≈ **62.7825** → 0.2625 → **₹ 26,250**; rebook at 1-month buying 62.6400 − 0.06264 = 62.57736 ≈ **₹ 62.5775**. **TYK P32 (import $ 20,000 at 42.32, extended 3 months):** cancel at spot buying 41.5000 − 0.075% = 41.4689 ≈ 41.47 → 8,46,400 − 8,29,400 = **₹ 17,000**; new rate 41.5200 + 0.93% (0.3861) = 41.9061 + 0.20% (0.0838) = 41.9899 ≈ **₹ 41.99**.

Illus 8 (purchase $ 50,000 at 59.6000, on 5 May extended to 5 July): cancel at June forward selling 59.2425 + 0.0592 = 59.3017 ≈ 59.3025 → 0.2975 × 50,000 = **₹ 14,875 payable to the customer**; rebook at July buying 59.6300 − 0.0596 = 59.5704 ≈ **₹ 59.5700**.

Note (app-written): Illus 8 says "rebooked with the delivery 15th July"; the request and rate used are for 5th July.

Automatic cancellation (FEDAI Rule 6): an overdue forward contract with no instructions on or before due date is **automatically cancelled within 3 working days** after maturity. Customer pays any exchange difference but is **not entitled to any profit**. Late delivery → fresh contract at spot; extension after due date → forward rate for the desired period. Charges:

- → (i) **Exchange difference**: spot rate of offsetting position on the cancellation date (or 3 working days, **whichever is earlier**) vs contract rate.
- → (ii) **Swap loss**: bank offsets at spot on the due date and takes the opposite position at the earliest forward.
- → (iii) **Interest on outlay of funds**: (bank's interbank cover rate − spot of opposite position on due date) × FC amount, for the period from due date to actual cancellation or 3 working days, **whichever is later**.
- → Any profit to the bank is **not passed on** to the customer.

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Illus 9 (importer $ 2,00,000 at 64.4000, cover 64.2800, due 10 June, extension asked 14 June to 10 Aug; margin 0.10%, interest 12%): (i) cancellation rate = spot buying 13 June 63.6800 − 0.0636 = 63.6163 ≈ **63.6175**; (ii) 1,28,80,000 − 1,27,23,500 = **₹ 1,56,500**; (iii) swap on 10 June: sell spot 63.8000, buy June forward 63.9500 → 0.15 → **₹ 30,000**; (iv) outlay 64.2800 − 63.8000 = 0.48 × 2,00,000 = ₹ 96,000 × 12% for 3 days = **₹ 96**; (v) new rate August selling 64.2500 + 0.0643 = 64.3143 ≈ **₹ 64.3150**; (vi) total cost **₹ 1,86,596**.

Note (app-written): Illus 9 — ₹ 96,000 × 12% × 3/365 = ₹ 94.68; ICAI's ₹ 96 equals a 360-day year. Also 63.6800 − 0.0636 is 63.6164 (ICAI prints 63.6163 using the unrounded margin 0.06368); the rounded rate 63.6175 is unaffected.

12.3 Non-deliverable Forward Contract (NDF)

- → **Cash-settled**, short-term forward on a **thinly traded or non-convertible** currency; P/L = (agreed rate − spot at settlement) × notional amount. Has a **fixing date** (difference calculated) and **settlement date** (difference paid). Usually 1 month to 1 year, quoted and settled in **US$**.

12.4 Rollover of Deliverable Forward Contract

→ Exporter has no FC / importer has no local currency to deliver at maturity → **spot cancellation + new contract for a later date**.

→ **Rationale:** non-receipt of FC from client (export); shortage of local currency (import); non-agreement of payment with clients; forwards normally available only up to **one year**, so longer exposures use rollovers.

13. Futures Contracts

→ Like a forward in obligation and payoff, but: **standardised** terms (quantity/quality, time/place of delivery, payment method); **daily marking to market** → credit risk greatly reduced; **smaller contract sizes** → wider market. Bought/sold through a **broker**; price fixed at outset.

→ **Hedge working** (SM answers): no. of contracts = exposure ÷ contract size (to the nearest whole number as asked); position = receiver of FC sells FC futures (TYK P34) — or, in a quote per unit of home currency, buys home-currency futures (TYK P37, P41); futures gain/loss + actual spot conversion = net; interest on initial margin is a cost (TYK P25).

Feature	Forward	Futures
Amount	Flexible	Standard amount
Maturity	Any valid business date agreed	Standard date (e.g. second Tuesday of every month)
Furthest maturity	Open	12 months forward
Currencies	All currencies	Majors
Cross rates	Available in one contract	Usually two contracts
Market-place	Global network	Regular futures markets / exchanges
Price fluctuations	No daily limit in many currencies	Daily price limit set by exchange
Risk	Depends on counter party	Minimal due to margin requirements
Honouring	Taking and giving delivery	Mostly by a reverse transaction
Cash flow	None until maturity	Initial margin + variation margin (MTM) + final payment
Trading hours	24 hours a day	4–8 hour sessions

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TYK P34 (ABC Technologic, $ 4,00,000): sell 400 contracts of $ 1,000 at 45.00, close at 44.50 → gain ₹ 2,00,000 + 44.50 × 4,00,000 = ₹ 1,80,00,000 → effective **₹ 45 per $**.

TYK P25 (EFD, $ 1 crore): forward $ 10,000,000 ÷ 0.016129 = ₹ 620,001,240; futures 25 contracts, margin 25 × 22,500 = ₹ 5,62,500; spot receipt ÷ 0.016136 = ₹ 619,732,276 + variation margin (0.000016 × 25 × 24,816,975)/0.016136 = ₹ 615,195 − interest 5,62,500 × 7% × 3/12 = ₹ 9,844 → **₹ 620,337,627** → futures best (no hedge ₹ 619,732,276).

TYK P20 (IMM arbitrage): buy £ 62,500 futures at 1.2806 ($ 80,037.50), sell forward at 1.2816 ($ 80,100) → profit **$ 62.50**.

TYK P37 (2az, € 2.8 m): (a) invoice in £ at 1.1965 → **£ 2.340 m**; (b) forward 1.1970 − 0.0055 = 1.1915 → **£ 2.35 m**; (c) 2.8 m ÷ 1.1943 = £ 2,344,470 → 37 contracts; buy at 1.1943, sell at 1.1873 → loss 37 × 62,500 × 0.0070 = € 16,188; € 27,83,812 ÷ 1.1873 = £ 2.3446 m → **(b) preferred** (also doubt whether German buyer accepts £ invoice).

TYK P41 (Nitrogen, € 4 m): (i) ÷ 1.1770 = £ 33,98,471; (ii) forward 1.1770 + 0.0060 = 1.1830 → £ 33,81,234; (iii) ÷ 1.1760 = £ 34,01,360 → 54 contracts; gain 54 × 62,500 × 0.0025 = € 8,438; € 40,08,438 ÷ 1.1785 = **£ 34,01,305** → **(iii) preferred**.

Note (app-written): TYK P37 — £ 2,344,470 ÷ 62,500 = 37.51, which rounds to 38 to the nearest whole number; ICAI uses 37 contracts. Keep ICAI's 37.

14. Option Contracts

→ **Call**: right, not obligation, to **buy** a specified quantity of a commodity/currency at a fixed price on or up to a date. **Put**: right to **sell**. Buyer = holder; seller = writer/grantor. Premium may be a % of the price per unit.

→ **American**: exercise any time in the period; **European**: only at the end. Holder may exercise, let it lapse or sell the option.

→ **Hedge working**: importer of FC buys a **call on FC** (or a put on home currency when the quote is FC per home — TYK P33; a put on £ for a UK importer of $ — TYK P40); contracts = exposure ÷ contract size (whole contracts), **exercise only if favourable**, premium paid upfront (convert at spot; add interest on premium if asked), uncovered balance at forward.

Point	Options	Futures
(a) Obligation	Only the seller (writer)	Both parties
(b) Premium	Paid by buyer to seller at inception	None
(c) Gain/loss	Buyer: loss restricted, gain unlimited	Unlimited gain/loss for buyer
(d) Exercise	American: any time during the period	Honoured only on the specified date

Price	Call buyer	Long futures	Call seller	Put buyer	Short futures	Put seller
Rises	Unlimited gain	Unlimited gain	Unlimited loss	Limited loss	Unlimited loss	Limited gain
Falls	Limited loss	Unlimited loss*	Limited gain	Unlimited gain*	Unlimited gain*	Unlimited loss*
Unchanged	Limited loss	No gain/loss	Limited gain	Limited loss	No gain/loss	Limited gain

* Price cannot go below zero, so technically limited. Transaction costs ignored.

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TYK P33 (XYZ, JY 5,00,000): forward 1/1.9726 = ₹ 0.5070 → **₹ 2,53,500**; put on INR at JY 2.125: 5,00,000 ÷ 2.125 = ₹ 2,35,294 + premium (2,35,294 × 0.098 ÷ 1.9516) ₹ 11,815 = **₹ 2,47,109** → option.

TYK P36 (Can$ payments): forward July 10,10,000 × 0.9501 = $ 9,59,601; Sept 7,05,000 × 0.9556 = $ 6,73,698. Options: 20 contracts at 0.94 = $ 9,40,000 + balance $ 9,501 + premium 50,000 × 20 × 0.0102 = $ 10,200 → $ 9,59,701; 14 at 0.95 = $ 6,65,000 + $ 4,778 + 50,000 × 14 × 0.0164 = $ 11,480 → $ 6,81,258 → **forward** (cheapest, firm risk averse).

TYK P39 (XYZ US, £ 3,00,000): (a) forward × 1.96 = **$ 5,88,000**; (b) MMH $ 6,05,741; (c) call 1.97 + 0.04: expected $ 5,94,900 + interest on premium 12,000 × 5.5% = 660 → $ 5,95,560; (d) no hedge Σ = **$ 5,86,500** → ICAI prefers **no hedge**.

TYK P40 (A Ltd UK, $ 3,64,897): forward × 0.64704 = £ 2,36,103; MMH £ 2,36,510; option: 17 puts on £ 12,500 at 1.70 = $ 3,61,250, balance $ 3,647 × 0.64704 = £ 2,360, premium 17 × 12,500 × 0.096 = $ 20,400 × 0.64033 = £ 13,063 → £ 2,12,500 + 13,063 + 2,360 = **£ 2,27,923** → option.

15. Swap Contracts

→ "Infinitely flexible": exchange the underlying economic basis of a debt or asset **without affecting the principal obligation**. Participants exchange cash flows at **payment/settlement dates** (fixed, or quantity × reference rate).

→ Groups: **interest rate, currency, commodity, equity**. Notional principal (except currency swaps) is used only to compute payments, **not exchanged**; interim payments usually **netted**. Users: large banks and corporations; swaps create credit exposure.

15.1–15.2 Interest Rate Swaps; Currency Swaps

→ **15.1 IRS**: covered in Chapter 12 of the SM.

→ **15.2 Currency swap** = exchange of liabilities between currencies. Three stages: (1) **spot exchange of principal**; (2) **continuing exchange of interest** — a series of forwards, usually at the initial spot rate; (3) **re-exchange of principal** on maturity.

→ **Benefits:** hedge currency risk; **cost savings** (strong Yen borrower raises in Tokyo and swaps into $); access to currencies that would otherwise command a high premium; **diversification** of borrowings.

→ **Currency coupon swap (Circus swap)**: fixed or floating interest in one currency for floating in another. Principal exchanged at the start, at the end, both, or neither. Often linked to a **Eurobond** issue — issue where cheapest, swap into the required currency via the underwriter.

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TYK PSO (Drilldip, ₹ 500 crore in, ₹ 740 crore back): swap — buy ₹ 500 cr at ₹ 50 (US$ 100 m), swap back at ₹ 50 = $ 100 m + ₹ 240 cr at ₹ 54 = $ 44.44 m − interest 8% $ 8 m → net **US$ 36.44 m**; without swap ₹ 740 cr ÷ 54 = $ 137.04 m − 8 → **US$ 29.04 m** → opt for swap.

15.3–15.4 Commodity Swaps; Equity Swaps

→ **Commodity swap**: series of futures-like settlements on a notional amount over multiple dates at specified reference prices/indices; mainly for **oil price risk**.

→ **Equity swap**: **total return** on equity or an index (dividend + capital) exchanged for a fixed or floating interest rate.

16. Popular Derivative Products (16.1–16.5)

Product	What / risk	SM example (notional USD 5,00,000)
16.1 Forward Importer	Counterparty **buys** USD at forward rate. Protected if settlement rate > forward; if lower, buys worse than market — loss potentially unlimited.	Strike 80: at 75 loss ₹ 25,00,000; 78.50 loss ₹ 7,50,000; 82.50 gain ₹ 12,50,000
16.2 Forward Exporter	Counterparty **sells** USD at forward. Protected against USD depreciation; if rate > forward, potentially unlimited loss.	Strike 80: 75 gain ₹ 25,00,000; 78.50 gain ₹ 7,50,000; 82.50 loss ₹ 12,50,000
16.3 FX Call Option	Importer pays premium; right to buy at strike if expiry rate ≥ strike; else buys at market (keeps INR appreciation).	Strike 78: 64.0080, 71.1200 no settlement; 78.2320 → gain ₹ 1,16,000
16.4 FX Put Option	Exporter pays premium; right to sell at strike if expiry rate ≤ strike; else sells at market (keeps USD appreciation).	Strike 78: 75.00 → gain ₹ 15,00,000; 79.00 no settlement
16.5 IRS USD floating to fixed	Counterparty receives floating USD, pays fixed → floating exposure becomes fixed. Floating < fixed on sustained basis → net negative; > fixed → net positive.	Fixed 1.50% p.a. semi-annual on USD 10,00,000 = pays 7,500: floating 0% → (7,500); 0.15% receives 750 → (6,750); 2% receives 10,000 → 2,500

17. Strategies for Exposure Management (17.1–17.4)

→ Shaped by attitude to risk, financial strength, nature of business, vulnerability; no single strategy fits all. Four options:

→ **17.1 Low risk – low reward**: **automatic hedging** in the forward market as soon as exposures arise. Merits: known yields/costs, little cash-flow destabilisation, no managerial time. Demerit: hardly optimum costs — not an optimum strategy (currencies behave like commodities).

→ **17.2 Low risk – reasonable reward**: **selective hedging** when forward rates are attractive, else keep open; needs quantified expectations; like a bond (hedged = known) and equity (unhedged = uncertain) mix.

→ **17.3 High risk – low reward**: leave **all exposures unhedged** — perhaps the worst; very high cash-flow destabilisation; only merit zero managerial effort.

→ **17.4 High risk – high reward**: **active trading** through continuous cancellations and re-bookings; trading becomes a **profit centre**; adopt only in full consciousness of risks.

18. Conclusion

→ Globalisation gives better international allocation of capital and diversification, but brings exchange-rate risk, political risk and interdependence. Rising rate volatility, trade, global markets and MNCs make study of these risks and their reduction imperative.

Exam steps — which rate: write each quote as "home per foreign" (direct) → customer sells FC = bid, buys FC = ask (bank's view: buys at bid, sells at ask) → indirect quote: pick the side that gives the customer less → margin: deduct from buying, add to selling → round as told (0.0025 / 0.0001 / whole rupee).

Exam steps — forward rate from points / broken date: ascending points = premium (add), descending = discount (deduct) → points × 0.0001 → for an odd date, interpolate pro rata (points for full period × days ÷ full days, or add the share of the next month's increment) → add margin → annualise [(Forward Rate − Spot rate)/(Spot rate)] x 12/months x 100.

Exam steps — cross rate / arbitrage: chain ₹/$ x $/¥ = ₹/¥ (bid × bid, ask × ask; divide where a leg is inverted) → compare with the quoted rate → route: buy in the cheaper market, sell in the dearer → end amount − start = gain.

Exam steps — IRP arbitrage: $(1 + r_D) = {}^{F}/_{S}\ (1 + r_F)$ with period rates → annualised premium vs interest differential → borrow the currency that is "cheap" after the forward, convert spot, invest, convert back at forward, repay loan + interest → gain.

Exam steps — hedge comparison (exporter / importer): forward (amount × forward) → MMH (PV at FC rate → spot → compound at home rate) → futures (contracts = exposure ÷ size; futures P/L + spot conversion − margin interest) → options (whole contracts, exercise if favourable, premium at spot + interest, balance at forward) → unhedged (expected spot = Σ rate × probability) → receivable: pick highest; payable: pick lowest.

Exam steps — fate of forward contract: identify bank's side (sale to importer / purchase from exporter) → reverse at opposite rate: spot (on due date), forward for original date (early), spot on cancellation date (after due date, auto-cancel in 3 working days) ± margin → difference × amount + flat charges → early delivery / late: add swap loss (spot vs forward reversal of cover) + interest on outlay (PLR / given rate × days) → extension: rebook at forward for new date ± margin → late: profit to bank not passed on.

Exam steps — import payment options: pay now = FC × spot (ask) + borrowing interest to due date; pay later = (FC + supplier interest) × forward (ask); gap periods add bank-loan interest → lowest rupee outflow wins.

Exam steps — exchange and cash position: list purchases vs sales (all deals, incl. forwards, bills, cancellations) → overbought / oversold → Nostro: only ready deliveries (TT remitted / received) → spot deal brings Nostro to target (also moves position) → forward deal sets the required overbought/oversold position.

Past papers

⑲ Foreign Exchange Exposure and Risk Management — what ICAI asked (Sep 2025, Jan 2026, May 2026)

41 marks set across the last three attempts (11.6% of marks set). Question labels and marks are ICAI's; the one-line concept descriptions are app-written.

Attempt	Question	Marks	Concept tested
Sep 2025	MCQ CS-IV 10-12	6	Forward quotes from swap points: interpolating swap points for a broken date (2 months 15 days), forward bid/ask, annualised premium/discount (MCQ)
Sep 2025	Q1(b)	4	Importer's yen payable: expected loss if left unhedged (expected spot after a 10% decline) versus forward cover; recommendation (Numerical)
Sep 2025	Q6(b)	7	Receivable hedge for a UK company: money-market hedge (borrow now) versus 12-month forward; net GBP proceeds and advice (Numerical)
Jan 2026	MCQ CS-II 4-6	6	Early delivery of a forward sale contract: swap loss, interest on outlay of funds, net conversion rate realised (MCQ)
Jan 2026	Q2(a)	6	Import payment choices: pay now with 1% discount (cash plus loan) versus 60-day or 90-day credit with interest, using forward and spot rates; cheapest option (Numerical)
Jan 2026	Q3(b)	4	Cross rates (bid and ask) from two direct quotes and arbitrage test through direct versus cross-rate route (Numerical)
May 2026	MCQ CS-V 13-15	6	Forward purchase contract: maturity settlement cost, closing out and rebooking, early settlement outflow, mark-to-market gain/loss on cancellation (MCQ)
May 2026	Q5(c)	2	Why transaction exposure is generally more critical than translation exposure (Theory)

What follows, exactly as ICAI printed it in the AFM study material: (1) Test Your Knowledge theory questions, with the SM paragraph ICAI's answer points to; (2) every SM illustration with ICAI's solution, grouped by SM section; (3) Test Your Knowledge practical questions with ICAI's solutions.

Theory questions (SM Test Your Knowledge) with ICAI answers

Theory Q1

1. "Operations in foreign exchange are exposed to a number of risks." Discuss.

ICAI answer: "Please refer paragraph 10" – SM paragraph 10 reproduced

10. FOREIGN EXCHANGE EXPOSURE

"An Exposure can be defined as a Contracted, Projected or Contingent Cash Flow whose magnitude is not certain at the moment. The magnitude depends on the value of variables such as Foreign Exchange rates and Interest rates."

Accordingly Foreign Exchange exposure refers to those parts of a company's business that would be affected if exchange rate changes. Foreign exchange exposures arise from many different activities.

For example, travellers going to visit another country have the risk that if that country's currency appreciates against their own their trip will be more expensive.

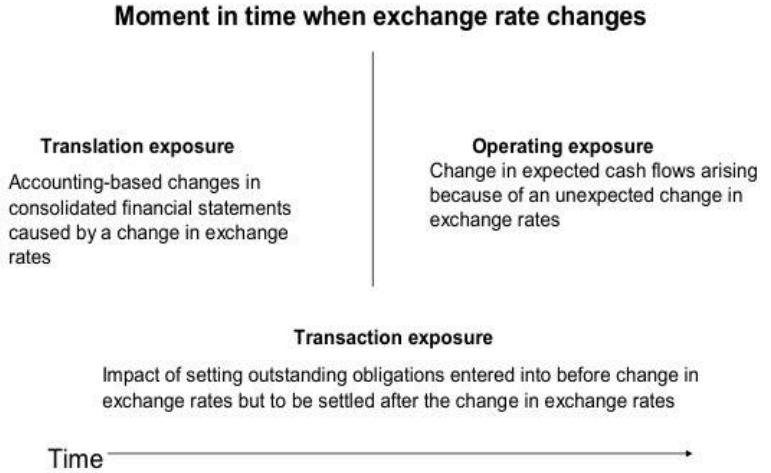
An exporter who sells his product in foreign currency has the risk that if the value of that foreign currency falls then the revenues in the exporter's home currency will be lower.

An importer who buys goods priced in foreign currency has the risk that the foreign currency will appreciate thereby making the local currency cost greater than expected.

Fund Managers and companies who own foreign assets are exposed to fall in the currencies where they own the assets. This is because if they were to sell those assets their exchange rate would have a negative effect on the home currency value.

Other foreign exchange exposures are less obvious and relate to the exporting and importing in ones local currency but where exchange rate movements are affecting the negotiated price.

The foreign exchange exposure may be classified under three broad categories:



10.1 Transaction Exposure

It measures the effect of an exchange rate change on outstanding obligations that existed before exchange rates changed but were settled after the exchange rate changes. Thus, it deals with cash flows that result from existing contractual obligations.

Example: If an Indian exporter has a receivable of $100,000 due in six months hence and if the dollar depreciates relative, to the rupee a cash loss occurs. Conversely, if the dollar appreciates relative to the rupee, a cash gain occurs.

The above example illustrates that whenever a firm has foreign currency denominated receivables or payables, it is subject to transaction exposure and their settlements will affect the firm's cash flow position.

It measures the changes in the value of outstanding financial obligation incurred prior to a change in exchange rates but not due to be settled until after the exchange rates change.

Thus, it deals with the changes in the cashflow which arise from existing contractual obligation.

In fact, the transaction exposures are the most common ones amongst all the exposures. Let's take an example of a company which exports to US and the export receivables are also denominated in USD. While doing budgeting the company had assumed rate of ₹ 62 per USD. By the time the exchange inward remittance arrives USD/INR could move down to ₹ 57 leading to wiping off of commercial profit for exporter. Such transaction exposures arise whenever a business has foreign currency denominated receipts or payments. The risk is an adverse movement of the exchange rate from the time the transaction is budgeted till the time the exposure is extinguished by sale or purchase of the foreign currency against the domestic currency.

10.2 Translation Exposure

It refers to gains or losses caused by the translation of foreign currency assets and liabilities into the currency of the parent company for consolidation purposes.

Translation exposure, also called as accounting exposure, is the potential for accounting derived changes in owner's equity to occur because of the need to "translate" foreign currency financial statements of foreign subsidiaries into a single reporting currency to prepare worldwide consolidated financial statements.

Translation exposures arise due to the need to "translate" foreign currency assets and liabilities into the home currency for the purpose of finalizing the accounts for any given period. A typical example of translation exposure is the treatment of foreign currency loans.

Consider that a company has taken a medium term loan to finance the import of capital goods worth dollars 1 million. When the import materialized, the exchange rate was, say, USD/INRR-55. The imported fixed asset was, therefore, capitalized in the books of the company at ₹ 550 lacs through the following accounting entry:

Debit fixed assets ₹ 550 lacs

Credit dollar loan ₹ 550 lacs

In the ordinary course assuming no change in the exchange rate, the company would have provided depreciation on the asset valued at ₹ 550 lacs, for finalizing its account for the year in which the asset was purchased.

However, what happens if at the time of finalization of the accounts the exchange rate has moved to say USD/INR-58. Now the dollar loan will have to be "translated" at ₹ 58, involving a "translation loss" of a ₹ 30 lacs. It shall have to be capitalized by increasing the book value of the asset, thus making the figure ₹ 580 lacs and consequently higher depreciation will have to be provided, thus reducing the net profit.

It will be readily seen that both transaction and translation exposures affect the bottom line of a company. The effect could be positive as well if the movement is favourable – i.e., in the cited examples, in case the USD would have appreciated and the USD would have depreciated against the rupee.

An important observation is that the translation exposure, of course, becomes a transaction exposure at some stage: the dollar loan has to be repaid by undertaking the transaction of purchasing dollars.

10.3 Economic Exposure

It refers to the extent to which the economic value of a company can decline due to changes in exchange rate. It is the overall impact of exchange rate changes on the value of the firm. The essence of economic exposure is that exchange rate changes significantly alter the cost of a firm's inputs and the prices of its outputs and thereby influence its competitive position substantially.

Effects of Local Currency Fluctuations on Company's Economic Exposure (Cash inflow)

Variables influencing the inflow of cash in Local currency	*Revaluation impact*	*Devaluation impact*
Local sale, relative to foreign competition in local currency	Decrease	Increase
Company's export in local currency	Decrease	Increase
Company's export in foreign currency	Decrease	Increase
Interest payments from foreign investments	Decrease	Increase

Effects of Local Currency Fluctuations on Company's Economic Exposure (Cash outflow)

Variables influencing the outflow of cash in local currency	*Revaluation impact*	*Devaluation impact*
Company's import of material the same denoted in local currency	Remain the same	Remain the same
Company's import of material denoted in foreign currency	Decrease	Increase
Interest on foreign debt	Decrease	Increase

Foreign exchange risk management is a critical component of any business that deals with international transactions. It involves the identification, assessment, and mitigation of risks associated with fluctuations in foreign currency exchange rates. The importance of foreign exchange risk management cannot be overstated, and here are some reasons why:

(i) *Protection against volatility:* Exchange rates are highly volatile and can change rapidly, which can result in significant losses for a business. Foreign exchange risk management helps to protect against this volatility by allowing businesses to lock in exchange rates in advance, providing greater stability and certainty in financial planning.

(ii) *Cost reduction:* Effective foreign exchange risk management can help businesses reduce costs associated with foreign transactions. By minimizing currency exchange rate losses and reducing the need for hedging, businesses can save significant amounts of money in the long run.

(iii) *Competitive advantage:* Companies that effectively manage their foreign exchange risks can gain a competitive advantage over their competitors. They can offer more competitive prices and more attractive payment terms, which can help to attract and retain customers.

(iv) *Improved cash flow:* Foreign exchange risk management can also help businesses to improve their cash flow by providing greater visibility and predictability in their international transactions. This can help businesses to better manage their cash flow and ensure that they have sufficient funds to meet their obligations.

(v) *Compliance with regulations:* Many countries have regulations in place that require businesses to manage their foreign exchange risks. Failure to comply with these regulations can result in significant fines and penalties. Effective foreign exchange risk management can help businesses to stay in compliance with these regulations and avoid potential legal issues.

In summary, foreign exchange risk management is critical for businesses that engage in international transactions. It helps to protect against volatility, reduce costs, gain a competitive advantage, improve cash flow, and ensure compliance with regulations. By managing foreign exchange risks effectively, businesses can achieve greater financial stability and success in the global marketplace.

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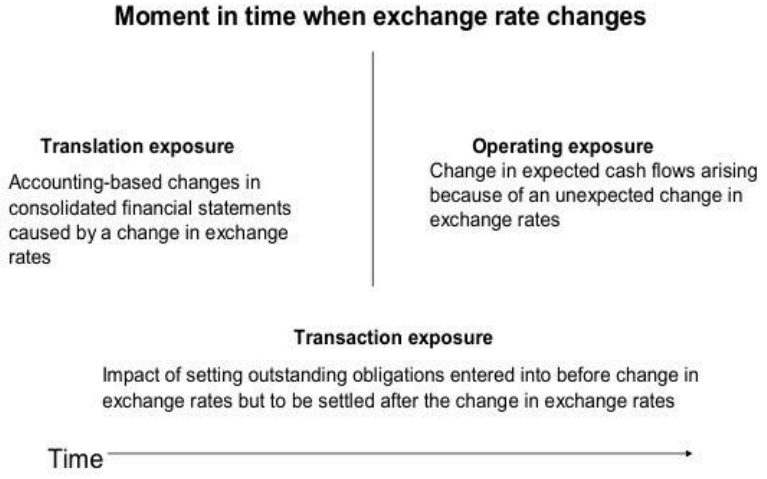
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In summary, foreign exchange risk management is critical for businesses that engage in international transactions. It helps to protect against volatility, reduce costs, gain a competitive advantage, improve cash flow, and ensure compliance with regulations. By managing foreign exchange risks effectively, businesses can achieve greater financial stability and success in the global marketplace.

Theory Q2

2. What do you mean by Nostro, Vostro and Loro Accounts?

ICAI answer: "Please refer paragraph 4" – SM paragraph 4 reproduced

4. NOSTRO, VOSTRO AND LORO ACCOUNTS

In interbank transactions, foreign exchange is transferred from one account to another account and from one centre to another centre. Therefore, the banks maintain three types of current accounts in order to facilitate quick transfer of funds in different currencies. These accounts are Nostro, Vostro and Loro accounts meaning "our", "your" and "their". A bank's foreign currency account maintained by the bank in a foreign country and in the home currency of that country is known as Nostro Account or "our account with you". For example, An Indian bank's Swiss franc account with a bank in Switzerland.

Vostro account is the account which is held by a foreign bank with a local bank, so if Citibank maintains an account with State bank of India it will be a Vostro account for State bank of India. The account which is Nostro for one bank is Vostro for another.

When domestic banks use the account of third party banks which holds a Nostro account to settle foreign exchange transactions these type of transactions are included under the Loro Account e.g. HDFC Bank has an account with Citi Bank but IDBI Bank doesn't have any Nostro account with Citi Bank. IDBI Bank has to pay the bill of imported machinery from the USA on behalf of its customers IDBI Bank approaches to HDFC Bank and ask to settle the invoice on its own behalf. HDFC bank worked as an intermediary between Citi Bank and IDBI Bank.

4.1 Exchange Position

It is referred to as total of purchase and sale commitments of a bank to purchase or sale foreign exchange whether actual delivery has taken place or not. In other words, all transactions for which bank has agreed with counter party are entered into exchange position on the date of the contract.

4.2 Cash Position

It is outstanding balance (debit or credit) in bank's Nostro account. Since all foreign exchange dealings of bank are routed through Nostro account it is credited for all purchases and debited for sales by the bank.

It should however be noted that all dealings whether delivery has taken place or not effects the Exchange Position but Cash Position is effected only when actual delivery has taken place.

Therefore, all transactions affecting Cash position will affect Exchange Position not vice versa.

Illustration 2

Suppose you are a dealer of ABC Bank and on 20.10.2014 you found that balance in your Nostro account with XYZ Bank in London is £65,000 and you had overbought £35,000. During the day following transaction have taken place:

	£
DD purchased	12,500
Purchased a Bill on London	40,000
Sold forward TT	30,000
Forward purchase contract cancelled	15,000
Remitted by TT	37,500
Draft on London cancelled	15,000

What steps would you take, if you are required to maintain a credit Balance of £7,500 in the Nostro A/c and keep as overbought position on £7,500?

Solution

Exchange Position:

Particulars	Purchase £	Sale £
Opening Balance Overbought	35,000	—
DD Purchased	12,500	—
Purchased a Bill on London	40,000	—
Sold forward TT	—	30,000
Forward purchase contract cancelled	—	15,000
TT Remittance		37,500
Draft on London cancelled	15,000	—
	1,02,500	82,500
Closing Balance Overbougt	—	20,000
	1,02,500	1,02,500

Cash Position (Nostro A/c)

	Credit £	Debit £
Opening balance credit	65,000	—
TT Remittance	—	37,500
	65,000	37,500
Closing balance (credit)	—	27,500
	65,000	65,000

To maintain Cash Balance in Nostro Account at £7,500 you have to sell £20,000 in Spot which will bring Overbought exchange position to Nil. Since bank require Overbought position of £7,500 it has to buy the same in forward market.

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① SM illustrations with ICAI solutions

1. Introduction

SM Ch 10 · Illustration 1

Illustration 1

If the Indian rupee is the home currency and the foreign currency is the US Dollar then what is the exchange rate between the rupee and the US dollar?

Solution — ICAI

Solution

- One can buy 0.0217 US dollars for one Indian rupee.
- ₹ 46.08 Indian rupees are needed to buy one US dollar.

(b) The Forward Market: A forward exchange contract occurs when buyers and sellers of currencies agree to deliver the currency at some future date. They agree to transact a specific amount of currency at a specific rate at a specified future date. The forward exchange rate is set and agreed by the parties and remains fixed for the contract period regardless of the fluctuations in the spot exchange rates in future. The forward exchange transactions can be understood by an example.

A US exporter of computer peripherals might sell computer peripherals to a German importer with immediate delivery but not require payment for 60 days. The German importer has an obligation to pay the required dollars in 60 days, so he may enter into a contract with a trader (typically a local banker) to deliver Euros for dollars in 60 days at a forward rate – the rate today for future delivery.

So, a forward exchange contract implies a forward delivery at specified future date of one currency for a specified amount of another currency. The exchange rate is agreed today, though the actual transactions of buying and selling will take place on the specified date only. The forward rate is not the same as the spot exchange rate that will prevail in future. The actual spot rate that may prevail on the specified date is not known today and only the forward rate for that day is known. The actual spot rate on that day will depend upon the supply and demand forces on that day. The actual spot rate on that day may be lower or higher than the forward rate agreed today.

An Indian exporter of goods to London could enter into a forward contract with his banker to sell pound sterling 90 days from now. This contract can also be described as a contract to purchase Indian Rupees in exchange for delivery of pound sterling. In other words, foreign exchange markets are the only markets where barter happens – i.e., money is delivered in exchange for money.

4. Nostro, Vostro And Loro Accounts

SM Ch 10 · Illustration 2 (under 4.2 Cash Position)

Illustration 2

Suppose you are a dealer of ABC Bank and on 20.10.2014 you found that balance in your Nostro account with XYZ Bank in London is £65,000 and you had overbought £35,000. During the day following transaction have taken place:

	£
DD purchased	12,500
Purchased a Bill on London	40,000
Sold forward TT	30,000
Forward purchase contract cancelled	15,000
Remitted by TT	37,500
Draft on London cancelled	15,000

What steps would you take, if you are required to maintain a credit Balance of £7,500 in the Nostro A/c and keep as overbought position on £7,500?

Solution — ICAI

Solution

Exchange Position:

Particulars	Purchase £	Sale £
Opening Balance Overbought	35,000	—
DD Purchased	12,500	—
Purchased a Bill on London	40,000	—
Sold forward TT	—	30,000
Forward purchase contract cancelled	—	15,000
TT Remittance		37,500
Draft on London cancelled	15,000	—
	1,02,500	82,500
Closing Balance Overbougt	—	20,000
	1,02,500	1,02,500

Cash Position (Nostro A/c)

	Credit £	Debit £
Opening balance credit	65,000	—
TT Remittance	—	37,500
	65,000	37,500
Closing balance (credit)	—	27,500
	65,000	65,000

To maintain Cash Balance in Nostro Account at £7,500 you have to sell £20,000 in Spot which will bring Overbought exchange position to Nil. Since bank require Overbought position of £7,500 it has to buy the same in forward market.

12. Forward Contract

SM Ch 10 · Illustration 3 (under 12.2 Fate of Forward Contracts)

Illustration 3

On 1st June 2015 the bank enters into a forward contract for 2 months for selling US$ 1,00,000 at ₹ 65.5000. On 31st July 2015 the spot rate was ₹ 65.7500/65.2500. Calculate the amount to be debited in the customer's account.

Solution — ICAI

Solution

The bank will apply rate originally agreed upon i.e., ₹ 65.5000 and will debit the account of the customer with ₹ 65,50,000.

Early Delivery

The bank may accept the request of customer of delivery before due date of forward contract provided the customer is ready to bear the loss if any that may accrue to the bank as a result of this. In addition to some prescribed fixed charges bank may also charge additional charges comprising of:

(a) *Swap Difference*: This difference can be loss/ gain to the bank. This arises on account of offsetting its position earlier created by early delivery as bank normally covers itself against the position taken in the original forward contract.

(b) *Interest on Outlay of Funds*: It might be possible early delivery request of a customer may result in outlay of funds. In such bank shall charge from the customer at a rate not less than prime lending rate for the period of early delivery to the original due date. However, if there is an inflow of funds the bank at its discretion may pass on interest to the customer at the rate applicable to term deposits for the same period.

SM Ch 10 · Illustration 4 (under 12.2 Fate of Forward Contracts)

Illustration 4

On 1 October 2015 Mr. X an exporter enters into a forward contract with a BNP Bank to sell US$ 1,00,000 on 31 December 2015 at ₹ 65.40/$. However, due to the request of the importer, Mr. X received amount on 28 November 2015. Mr. X requested the bank the take delivery of the remittance on 30 November 2015 i.e., before due date. The inter-banking rates on 28 November 2015 was as follows:

Spot	₹ 65.22/65.27
One Month Premium	10/15

If bank agrees to take early delivery then what will be net inflow to Mr. X assuming that the prevailing prime lending rate is 18%.

Solution — ICAI

Solution

Bank will buy from customer at the agreed rate of ₹ 65.40. In addition to the same if bank will charge/ pay swap difference and interest on outlay funds.

(a)	Swap Difference	
	Bank Sells at Spot Rate on 28 November 2015	₹ 65.22
	Bank Buys at Forward Rate of 31 December 2015 (65.27 + 0.15)	₹ 65.42
	Swap Loss per US$	₹ 00.20
	Swap loss for US$ 1,00,000	₹ 20,000
(b)	Interest on Outlay Funds	
	On 28th November Bank sells at	₹ 65.22
	It buys from customer at	₹ 65.40
	Outlay of Funds per US$	₹ 00.18
	Interest on Outlay fund for US$ 1,00,000 for 31 days	₹ 275.00
	(US$100000 x 00.18 x 31/365 x 18%)	
(c)	Charges for early delivery	
	Swap loss	₹ 20,000.00
	Interest on Outlay fund for US$ 1,00,000 for 31 days	₹ 275.00
		₹ 20,275.00
(d)	Net Inflow to Mr. X	
	Amount received on sale (₹ 65.40 x 1,00,000)	₹ 65,40,000
	Less: Charges for early delivery payable to bank	(₹ 20,275)
		₹ 65,19,725

Late Delivery

In case of late delivery, current rate prevailing on such date of delivery shall be applied. However, before this delivery (execution) takes place the provisions of Automatic Cancellation (discussed later on) shall be applied.

Cancellation on Due Date

In case of cancellation on due date in addition to flat charges (if any) the difference between contracted rate and the cancellation rate (reverse action of original contract) is charged from/ paid to the customer.

SM Ch 10 · Illustration 5 (under 12.2 Fate of Forward Contracts)

Illustration 5

On 15th January 2015 you as a banker booked a forward contract for US$ 250000 for your import customer deliverable on 15th March 2015 at ₹ 65.3450. On due date customer request you to cancel the contract. On this date quotation for US$ in the inter-bank market is as follows:

Spot	₹ 65.2900/2975 per US$
Spot/ April	3000/ 3100
Spot/ May	6000/ 6100

Assuming that the flat charges for the cancellation is ₹ 100 and exchange margin is 0.10%, then determine the cancellation charges payable by the customer.

Solution — ICAI

Solution

Bank will sell to customer at the agreed rate of ₹ 65.3450.

Since this is sale contract by the bank the contract shall be cancelled at ready buying rate on the date of cancellation as follows:

Spot Buying Rate on 15 March 2015	₹ 65.2900
Less: Exchange Margin	₹ 0.0653
	₹ 65.2247
Rounded to ₹ 65.2250	
Dollars sold to customer at	₹ 65.3450
Dollars bought from customer	₹ 65.2250
Net amount payable by the customer per US$	₹ 0.1200
Amount payable by the customer	
Flat Charges	₹ 100.00
Cancellation Charges (₹ 0.12 x 250000)	₹ 30,000.00
	₹ 30,100.00

Early Cancellation

If a forward is required to be cancelled earlier than the due date of forward contract same shall be cancelled at opposite rate of original contract of the date that synchronizes with the date of original forward contract.

SM Ch 10 · Illustration 6 (under 12.2 Fate of Forward Contracts)

Illustration 6

You as a banker has entered into a 3 month's forward contract with your customer to purchase AUD 1,00,000 at the rate of ₹ 47.2500. However, after 2 months your customer comes to you and requests cancellation of the contract. On this date quotation for AUD in the market is as follows:

Spot	₹ 47.3000/3500 per AUD
1 month forward	₹ 47.4500/5200 per AUD

Determine the cancellation charges payable by the customer.

Solution — ICAI

Solution

The contract shall be cancelled at the 1 month forward sale rate of ₹ 47.5200 as follows:

AUD bought from customer under original forward contract at	₹ 47.2500
On cancellation it is sold to him at	₹ 47.5200
Net amount payable by customer per AUD	₹ 00.2700

Thus, total cancellation charges payable by the customer ₹ 27,000

Late Cancellation

In case of late cancellation of Forward Contract, the provisions of Automatic Cancellation (discussed later on) shall be applied.

Extension on Due Date

It might also be possible that an exporter is not able to export goods on the due date. Similarly, it might also be possible that an importer is not able to pay on due date. In both of these situations an extension of contract for selling and buying contract is warranted. Accordingly, if earlier contract is extended first, it shall be cancelled and rebooked for the new delivery period. In case extension is on due date it shall be cancelled at spot rate as like cancellation on due date (discussed earlier) and new contract shall be rebooked at the forward rate for the new delivery period.

SM Ch 10 · Illustration 7 (under 12.2 Fate of Forward Contracts)

Illustration 7

Suppose you are a banker and one of your export customers has booked a US$ 1,00,000 forward sale contract for 2 months with you at the rate of ₹ 62.5200 and simultaneously you covered yourself in the interbank market at ₹ 62.5900. However, on due date, after 2 months your customer comes to you and requests for cancellation of the contract and also requests for extension of the contract by one month. On this date quotation for US$ in the market was as follows:

Spot	₹ 62.6800/62.7200
1 month forward	₹ 62.6400/62.7400

Determine the extension charges payable by the customer assuming exchange margin of 0.10% on buying as well as selling.

Solution — ICAI

Solution

Cancellation

First the original contract shall be cancelled at Spot Selling Rate as follows:

US$/₹ Spot Selling Rate	₹ 62.7200
Add: Margin @ 0.10%	₹ 0.06272
	₹ 62.78272

Rounded off	₹ 62.7825
Bank buys US$ under original contract at	₹ 62.5200
Bank Sells at Spot Rate	₹ 62.7825
	₹ 0.2625

Thus, total cancellation charges payable by the customer for US$ 1,00,000 is ₹ 26,250.

Rebooking

Forward US$/₹ Buying Rate	₹ 62.6400
Less: Margin @ 0.10%	₹ 0.06264
Net amount payable by customer per US$	₹ 62.57736
Rounded off	₹ 62.5775

Extension before Due Date

In case any request to extend the contract is received before due date of maturity of forward contract, first the original contract would be cancelled at the relevant forward rate as in case of cancellation of contract before due date and shall be rebooked at the current forward rate of the forward period.

SM Ch 10 · Illustration 8 (under 12.2 Fate of Forward Contracts)

Illustration 8

Suppose you as a banker entered into a forward purchase contract for US$ 50,000 on 5th March with an export customer for 3 months at the rate of ₹ 59.6000. On the same day you also covered yourself in the market at ₹ 60.6025. However, on 5th May your customer comes to you and requests extension of the contract to 5thJuly. On this date (5th May) quotation for US$ in the market is as follows:

Spot	₹ 59.1300/1400 per US$
Spot/ 5th June	₹ 59.2300/2425 per US$
Spot/ 5thJuly	₹ 59.6300/6425 per US$

Assuming a margin 0.10% on buying and selling, determine the extension charges payable by the customer and the new rate quoted to the customer.

Note: Rates to be rounded off to 4 decimals in multiples of 0.0025.

Solution — ICAI

Solution

(a) Cancellation of Original Contract

The forward purchase contract shall be cancelled at the for the forward sale rate for delivery June.

Interbank forward selling rate	₹ 59.2425
Add: Exchange Margin	₹ 0.0592
Net amount payable by customer per US$	₹ 59.3017
Rounded off, the rate applicable is ₹ 59.3025	
Buying US$ under original contract at original rate	₹ 59.6000
Selling rate to cancel the contract	₹ 59.3025
Difference per US$	₹ 00.2975

Exchange difference for US$ 50,000 payable to the customer is ₹ 14,875.

(b) Rate for booking new contract

The forward contract shall be rebooked with the delivery 15th July as follows:

Forward buying rate (5th July)	₹ 59.6300
Less: Exchange Margin	₹ 0.0596
	₹ 59.5704

Rounded off to ₹ 59.5700

Late Extension

In case of late extension current rate prevailing on such date of delivery shall be applied. However, before this delivery the provisions of Automatic Cancellation (discussed later on) shall be applied.

Automatic Cancellation

As per FEDAI Rule 6, a forward contract which remains overdue without any instructions from the customers on or before due date shall stand automatically cancelled within 3 working days after the maturity date. Though customer is liable to pay the exchange difference arising there from but not entitled for the profit resulting from this cancellation.

For late delivery and extension after due date as mentioned above the contract shall be treated as fresh contract and appropriate rates prevailing on such date shall be applicable as mentioned below:

1. Late Delivery: In this case the relevant spot rate prevailing on the such date shall be applicable.

2. Extension after Due Date: In this case relevant forward rate for the period desired shall be applicable.

As mentioned earlier in both of above case cancellation charges shall be payable consisting of following:

(i) *Exchange Difference:* The difference between Spot Rate of offsetting position (cancellation rate) on the date of cancellation of contract after due date or 3 working days (whichever is earlier) and original rate contracted for.

(ii) *Swap Loss (Difference):* The loss arises on account of offsetting its position at the spot rate on the date of cancellation and taking opposite position of offsetting position at earliest forward rate.

(iii) *Interest on Outlay of Funds:* Interest on the difference between the rate entered by the bank in the interbank market and actual spot rate on the due date of contract of the opposite position multiplied by the amount of foreign currency amount involved. This interest shall be calculated for the period from the due date of maturity of the contract and the actual date of cancellation of the contract or 3 working days whichever is later.

Please note in above in any case there is profit by the bank on any course of action same shall not be passed on the customer as normally passed cancellation and extension on or before due dates.

SM Ch 10 · Illustration 9 (under 12.2 Fate of Forward Contracts)

Illustration 9

An importer booked a forward contract with his bank on 10th April for USD 2,00,000 due on 10th June @ ₹ 64.4000. The bank covered its position in the market at ₹ 64.2800.

The exchange rates for dollar in the interbank market on 10th June and 13th June were:

	10th June	13th June
Spot USD 1=	₹ 63.8000/8200	₹ 63.6800/7200
Spot/June	₹ 63.9200/9500	₹ 63.8000/8500
July	₹ 64.0500/0900	₹ 63.9300/9900
August	₹ 64.3000/3500	₹ 64.1800/2500
September	₹ 64.6000/6600	₹ 64.4800/5600

Exchange Margin 0.10% and interest on outlay of funds @ 12%. The importer requested on 14th June for extension of contract with due date on 10th August.

Rates to be rounded off to 4 decimals in multiples of 0.0025.

On 10th June, Bank Swaps by selling spot and buying one month forward.

Calculate:

(i) Cancellation rate

(ii) Amount payable on $ 2,00,000

(iii) Swap loss

(iv) Interest on outlay of funds, if any

(v) New contract rate

(vi) Total Cost

Solution — ICAI

Solution

(i) Cancellation Rate:

The forward sale contract shall be cancelled at Spot TT Purchase for $ prevailing on the date of cancellation as follows:

$/ ₹ Market Buying Rate	₹ 63.6800
Less: Exchange Margin @ 0.10%	₹ 0.0636
	₹ 63.6163

Rounded off to ₹ 63.6175

(ii) Amount payable on $ 2,00,000

Bank sells $2,00,000 @ ₹ 64.4000	₹ 1,28,80,000
Bank buys $2,00,000 @ ₹ 63.6175	₹ 1,27,23,500
Amount payable by customer	₹ 1,56,500

(iii) Swap Loss

On 10th June the bank does a swap sale of $ at market buying rate of ₹ 63.8000 and forward purchase for June at market selling rate of ₹ 63.9500.

Bank buys at	₹ 63.9500
Bank sells at	₹ 63.8000
Amount payable by customer	₹ 0.1500

Swap Loss for $ 2,00,000 is ₹ = ₹ 30,000

(iv) Interest on Outlay of Funds

On 10th June, the bank receives delivery under cover contract at ₹ 64.2800 and sell spot at ₹ 63.8000.

Bank buys at	₹ 64.2800
Bank sells at	₹ 63.8000
Amount payable by customer	₹ 0.4800

Outlay for $ 2,00,000 is ₹ 96,000

Interest on ₹ 96,000 @ 12% for 3 days ₹ 96

(v) New Contract Rate

The contract will be extended at current rate

$/ ₹ Market forward selling Rate for August	₹ 64.2500
Add: Exchange Margin @ 0.10%	₹ 0.0643
	₹ 64.3143

Rounded off to ₹ 64.3150

(vi) Total Cost

Cancellation Charges	₹ 1,56,500.00
Swap Loss	₹ 30,000.00
Interest	₹ 96.00
	₹ 1,86,596.00

Ⓟ Practical questions (SM Test Your Knowledge) with ICAI solutions

Practical Q1

1. The price of a bond just before a year of maturity is $ 5,000. Its redemption value is $ 5,250 at the end of the said period. Interest is $ 350 p.a. The Dollar appreciates by 2% during the said period. Calculate the rate of return.

Solution — ICAI

1. Here we can assume two cases (i) If investor is US investor then there will be no impact of appreciation in $. (ii) If investor is from any other nation other than US say Indian then there will be impact of $ appreciation on his returns.

First we shall compute return on bond which will be common for both investors.

$$\text{Return} = \frac{(\text{Price at end - Price at begining}) + \text{Interest}}{\text{Price at begining}}$$

$$= \frac{(5250 - 5000) + 350}{5000}$$

$$= \frac{250 + 350}{5000} = 0.12 \text{ say } 12\%$$

(i) For US investor the return shall be 12% and there will be no impact of appreciation in $.

(ii) If $ appreciate by 2% then return for non-US investor shall be:

(1+0.12)(1+0.02) -1 =1.12X1.02 - 1 = 0.1424 i.e. 14.24%

Practical Q2

2. ABN-Amro Bank, Amsterdam, wants to purchase ₹ 15 million against US$ for funding their Nostro account with Canara Bank, New Delhi. Assuming the inter-bank, rates of US$ is ₹ 51.3625/3700, what would be the rate Canara Bank would quote to ABN-Amro Bank? Further, if the deal is struck, what would be the equivalent US$ amount.

Solution — ICAI

2. Here Canara Bank shall buy US$ and credit ₹ to Vostro account of ABN-Amro Bank. Canara Bank's buying rate will be based on the Inter-bank Buying Rate (as this is the rate at which Canara Bank can sell US$ in the Interbank market)

Accordingly, the Interbank Buying Rate of US$ will be ₹ 51.3625 (lower of two) i.e. (1/51.3625) = $ 0.01947/₹

Equivalent of US$ for ₹ 15 million at this rate will be

$$= \frac{15,000,000}{51.3625} = \text{US\$ } 2,92,041.86$$

or = 15,000,000 x $ 0.01947 = US$ 2,92,050

Practical Q3

3. XYZ LTD Bank, Amsterdam, wants to purchase ₹ 25 million against £ for funding their Nostro account and they have credited LORO account with Bank of London, London.

Calculate the amount of £'s credited. Ongoing inter-bank rates are per $, ₹ 61.3625/3700 & per £, $ 1.5260/70.

Solution — ICAI

3. To purchase Rupee, XYZ LTD Bank shall first sell £ and purchase $ and then sell $ to purchase Rupee. Accordingly, rate $(£/₹)_{ask}$ shall be used.

The available rates are as follows:

$(\$/£)_{bid}$ = $1.5260

$(\$/£)_{ask}$ = $1.5270

$(₹/\$)_{bid}$ = ₹ 61.3625

$(₹/\$)_{ask}$ = ₹ 61.3700

From above available rates we can compute required rate as follows:

$(£/₹)_{ask}$ = $(£/\$)_{ask}$ x $(\$/₹)_{ask}$

= (1/1.5260) x (1/61.3625)

= £ 0.01068 or £ 0.0107

Thus, amount of £ to be credited

= ₹ 25,000,000 x £ 0.0107

= £ 267,500

Practical Q4

4. ABC Ltd. of UK has exported goods worth Can $ 5,00,000 receivable in 6 months. The exporter wants to hedge the receipt in the forward market. The following information is available:

Spot Exchange Rate	Can $ 2.5/£
Interest Rate in UK	12%
Interest Rate in Canada	15%

The forward rates truly reflect the interest rates differential. Find out the gain/loss to UK exporter if Can $ spot rates (i) declines 2%, (ii) gains 4% or (iii) remains unchanged over next 6 months.

Solution — ICAI

4. Forward Rate = $\frac{2.50\,(1+0.075)}{(1+0.060)}$ = Can$ 2.535/£

(i) If spot rate decline by 2%

Spot Rate = Can$ 2.50 x 1.02 = Can$ 2.55/£

	£
£ receipt as per Forward Rate (Can $ 5,00,000/ Can$ 2.535)	1,97,239
£ receipt as per Spot Rate (Can $ 5,00,000/ Can$ 2.55)	1,96,078
Gain due to forward contract	1,161

(ii) If spot rate gains by 4%

Spot Rate = Can$ 2.50 x 0.96 = Can$ 2.40/£

	£
£ receipt as per Forward Rate (Can $ 5,00,000/ Can$ 2.535)	1,97,239
£ receipt as per Spot Rate (Can $ 5,00,000/ Can$ 2.40)	2,08,333
Loss due to forward contract	11,094

(iii) If spot rate remains unchanged

	£
£ receipt as per Forward Rate (Can $ 5,00,000/ Can$ 2.535)	1,97,239
£ receipt as per Spot Rate (Can $ 5,00,000/ Can$ 2.50)	2,00,000
Loss due to forward contract	2,761

Practical QS

5. On April 3, 2016, a Bank quotes the following:

Spot exchange Rate (US $ 1)	INR 66.2525	INR 67.5945
2 months' swap points	70	90
3 months' swap points	160	186

In a spot transaction, delivery is made after two days.

Assume spot date as April 5, 2016.

Assume 1 swap point = 0.0001,

You are required to:

(i) ascertain swap points for 2 months and 15 days. (For June 20, 2016),

(ii) determine foreign exchange rate for June 20, 2016, and

(iii) compute the annual rate of premium/discount of US$ on INR, on an average rate.

Solution — ICAI

5. (i) Swap Points for 2 months and 15 days

	Bid	Ask
Swap Points for 2 months (a)	70	90
Swap Points for 3 months (b)	160	186
Swap Points for 30 days (c) = (b) – (a)	90	96
Swap Points for 15 days (d) = (c)/2	45	48
Swap Points for 2 months & 15 days (e) = (a) + (d)	115	138

(ii) Foreign Exchange Rates for 20th June 2016

	Bid	Ask
Spot Rate (a)	66.2525	67.5945
Swap Points for 2 months & 15 days (b)	0.0115	0.0138
	66.2640	67.6083

(iii) Annual Rate of Premium

	Bid	Ask
Spot Rate (a)	66.2525	67.5945
Foreign Exchange Rates for 20th June 2016 (b)	66.2640	67.6083
Premium (c)	0.0115	0.0138
Total (d) = (a) + (b)	132.5165	135.2028
Average (d) / 2	66.2583	67.6014
Premium	$\frac{0.0115}{66.2583} \times \frac{12}{2.5} \times 100$ = 0.0833%	$\frac{0.0138}{67.6014} \times \frac{12}{2.5} \times 100$ = 0.0980%

Practical Q6

6. JKL Ltd., an Indian company has an export exposure of JPY 10,000,000 receivable August 31, 2014. Japanese Yen (JPY) is not directly quoted against Indian Rupee.

The current spot rates are:

INR/US $ = ₹ 62.22

JPY/US$ = JPY 102.34

It is estimated that Japanese Yen will depreciate to 124 level and Indian Rupee to depreciate against US $ to ₹ 65.

Forward rates for August 2014 are

INR/US $ = ₹ 66.50

JPY/US$ = JPY 110.35

Required:

(i) Calculate the expected loss, if the hedging is not done. How the position will change, if the firm takes forward cover?

(ii) If the spot rates on August 31, 2014 are:

INR/US $= ₹ 66.25

JPY/US$ = JPY 110.85

Is the decision to take forward cover justified?

Solution — ICAI

6. Since the direct quote for ¥ and ₹ is not available it will be calculated by cross exchange rate as follows:

₹/$ x $/¥ = ₹/¥

62.22/102.34 = 0.6080

Spot rate on date of export 1¥ = ₹ 0.6080

Expected Rate of ¥ for August 2014 = ₹ 0.5242 (₹ 65/¥124)

Forward Rate of ¥ for August 2014 = ₹ 0.6026 (₹ 66.50/¥110.35)

(i) Calculation of expected loss without hedging

Value of export at the time of export (₹ 0.6080 x ¥10,000,000)	₹ 60,80,000
Estimated payment to be received on Aug. 2014 (₹ 0.5242 x ¥10,000,000)	₹ 52,42,000
Loss	₹ 8,38,000

Hedging of loss under Forward Cover

₹ Value of export at the time of export (₹ 0.6080 x ¥10,000,000)	₹ 60,80,000
Payment to be received under Forward Cover (₹ 0.6026 x ¥10,000,000)	₹ 60,26,000
Loss	₹ 54,000

By taking forward cover loss is reduced to ₹ 54,000.

(ii) Actual Rate of ¥ on August 2014 = ₹ 0.5977 (₹ 66.25/¥110.85)

Value of export at the time of export (₹ 0.6080 x ¥10,000,000)	₹ 60,80,000
Estimated payment to be received on Aug. 2014 (₹ 0.5977 x ¥10,000,000)	₹ 59,77,000
Loss	₹ 1,03,000

The decision to take forward cover is still justified.

Practical Q7

7. You sold Hong Kong Dollar 1,00,00,000 value spot to your customer at ₹ 5.70 & covered yourself in London market on the same day, when the exchange rates were

US$ 1	=	H.K.$ 7.5880	7.5920
Local inter bank market rates for US$ were			
Spot US$ 1	=	₹ 42.70	42.85

Calculate cover rate and ascertain the profit or loss in the transaction. Ignore brokerage.

Solution — ICAI

7. The bank (Dealer) covers itself by buying from the market at market selling rate.

Rupee – Dollar selling rate	= ₹ 42.85
Dollar – Hong Kong Dollar	= HK $ 7.5880
Rupee – Hong Kong cross rate	= ₹ 42.85 / 7.5880
	= ₹ 5.6471

Profit / Loss to the Bank

Amount received from customer (1 crore × 5.70)	₹ 5,70,00,000
Amount paid on cover deal (1 crore × 5.6471)	₹ 5,64,71,000
Profit to Bank	₹ 5,29,000

Practical Q8

8. You, a foreign exchange dealer of your bank, are informed that your bank has sold a T.T. on Copenhagen for Danish Kroner 10,00,000 at the rate of Danish Kroner 1 = ₹ 6.5150. You are required to cover the transaction either in London or New York market. The rates on that date are as under:

Mumbai-London	₹ 74.3000	₹ 74.3200
Mumbai-New York	₹ 49.2500	₹ 49.2625
London-Copenhagen	DKK 11.4200	DKK 11.4350
New York-Copenhagen	DKK 07.5670	DKK 07.5840

In which market will you cover the transaction, London or New York, and what will be the exchange profit or loss on the transaction? Ignore brokerages.

Solution — ICAI

8. Amount realized on selling Danish Kroner 10,00,000 at ₹ 6.5150 per Kroner = ₹ 65,15,000.

Cover at London:

Bank buys Danish Kroner at London at the market selling rate.

Pound sterling required for the purchase (DKK 10,00,000 ÷ DKK 11.4200) = GBP 87,565.67

Bank buys locally GBP 87,565.67 for the above purchase at the market selling rate of ₹ 74.3200.

The rupee cost will be = ₹ 65,07,88

Profit (₹ 65,15,000 - ₹ 65,07,881) = ₹ 7,119

Cover at New York:

Bank buys Kroners at New York at the market selling rate.

Dollars required for the purchase of Danish Kroner (DKK10,00,000 ÷ 7.5670) = USD 1,32,152.77

Bank buys locally USD 1,32,152.77 for the above purchase at the market selling rate of ₹ 49.2625.

The rupee cost will be = ₹ 65,10,176.

Profit (₹ 65,15,000 - ₹ 65,10,176) = ₹ 4,824

The transaction would be covered through London which gets the maximum profit of ₹ 7,119 or lower cover cost at London Market by (₹ 65,10,176 - ₹ 65,07,881) = ₹ 2,295

Practical Q9

9. On January 28, 2013 an importer customer requested a Bank to remit Singapore Dollar (SGD) 2,500,000 under an irrevocable Letter of Credit (LC). However, due to unavoidable factors, the Bank could effect the remittances only on February 4, 2013. The inter-bank market rates were as follows:

		January 28, 2013	February 4, 2013
US$ 1=		₹ 45.85/45.90	₹ 45.91/45.97
GBP £ 1	=	US$ 1.7840/1.7850	US$ 1.7765/1.7775
GBP £ 1	=	SGD 3.1575/3.1590	SGD 3. 1380/3.1390

The Bank wishes to retain an exchange margin of 0.125%

Required:

How much does the customer stand to gain or lose due to the delay?

(Note: Calculate the rate in multiples of 0.0001)

Solution — ICAI

9. On January 28, 2013 the importer customer requested to remit SGD 25 lakhs.

To consider sell rate for the bank:

US $	=	₹45.90
Pound 1	=	US$ 1.7850
Pound 1	=	SGD 3.1575
Therefore, SGD 1	=	₹ 45.90 * 1.7850 / SGD 3.1575
SGD 1	=	₹25.9482
Add: Exchange margin (0.125%)		₹ 0.0324
		₹ 25.9806

On February 4, 2013 the rates are

US $	=	₹ 45.97
Pound 1	=	US$ 1.7775
Pound 1	=	SGD 3.1380
Therefore, SGD 1	=	₹ 45.97 * 1.7775 / SGD 3.1380
SGD 1	=	₹ 26.0394
Add: Exchange margin (0.125%)		₹ 0.0325
		₹ 26.0719

Hence, loss to the importer

= SGD 25,00,000 (₹26.0719 – ₹25.9806)= ₹2,28,250

Practical Q10

10. Following are the details of cash inflows and outflows in foreign currency denominations of MNP Co. an Indian export firm, which have no foreign subsidiaries:

Currency	Inflow	Outflow	Spot rate	Forward rate
US $	4,00,00,000	2,00,00,000	48.01	48.82
French Franc (FFr)	2,00,00,000	80,00,000	7.45	8.12
U.K. £	3,00,00,000	2,00,00,000	75.57	75.98
Japanese Yen	1,50,00,000	2,50,00,000	3.20	2.40

(i) Determine the net exposure of each foreign currency in terms of Rupees.

(ii) Are any of the exposure positions offsetting to some extent?

Solution — ICAI

10. (i) Net exposure of each foreign currency in Rupees

	Inflow (Millions)	Outflow (Millions)	Net Inflow (Millions)	Spread	Net Exposure (Millions)
US$	40	20	20	0.81	16.20
FFr	20	8	12	0.67	8.04
UK£	30	20	10	0.41	4.10
Japan Yen	15	25	-10	-0.80	8.00

(ii) In Japanese Yen, the net exposure is payable, and the forward rate is quoted at a discount, effectively offsetting the position. Likewise, in the remaining currencies, the net exposures are in receivables, and the related currencies are at a premium, offsetting the positions in their respective currencies.

Practical Q11

11. The following 2-way quotes appear in the foreign exchange market:

	Spot	2-months forward
₹/US $	₹ 46.00/₹ 46.25	₹ 47.00/₹ 47.50

Required:

(i) How many US dollars should a firm sell to get ₹ 25 lakhs after 2 months?

(ii) How many Rupees is the firm required to pay to obtain US $ 2,00,000 in the spot market?

(iii) Assume the firm has US $ 69,000 in current account earning no interest. ROI on Rupee investment is 10% p.a. Should the firm encash the US $ now or 2 months later?

Solution — ICAI

11. (i) US $ required to get ₹ 25 lakhs after 2 months at the Rate of ₹ 47/$

$$\therefore \frac{₹\ 25,00,000}{₹\ 47} = \text{US \$ } 53191.489$$

(ii) ₹ required to get US$ 2,00,000 now at the rate of ₹ 46.25/$

∴ US $ 200,000 × ₹ 46.25 = ₹ 92,50,000

(iii) Encashing US $ 69000 Now Vs 2 month later

Proceed if we can encash in open mkt $ 69000 × ₹46 = ₹ 31,74,000

Opportunity gain

$= 31{,}74{,}000 \times \frac{10}{100} \times \frac{2}{12}$ ₹ 52,900

Likely sum at end of 2 months 32,26,900

Proceeds if we can encash by forward rate :

$ 69000 × ₹47.00 32,43,000

It is better to encash the proceeds after 2 months and get opportunity gain.

Practical Q12

12. Z Ltd. importing goods worth USD 2 million, requires 90 days to make the payment. The overseas supplier has offered a 60 days interest free credit period and for additional credit for 30 days an interest of 8% per annum.

The bankers of Z Ltd offer a 30 days loan at 10% per annum and their quote for foreign exchange is as follows:

	₹
Spot 1 USD	56.50
60 days forward for 1 USD	57.10
90 days forward for 1 USD	57.50

You are required to evaluate the following options:

(i) Pay the supplier in 60 days, or

(ii) Avail the supplier's offer of 90 days credit.

Solution — ICAI

12. (i) Pay the supplier in 60 days

If the payment is made to supplier in 60 days the applicable forward rate for 1 USD	₹ 57.10
Payment Due	USD 2,000,000
Outflow in Rupees (USD 2000000 × ₹ 57.10)	₹ 114,200,000
Add: Interest on loan for 30 days@10% p.a.	₹ 9,51,667
Total Outflow in ₹	₹ 11,51,51,667

(ii) Availing supplier's offer of 90 days credit

Amount Payable	USD 2,000,000
Add: Interest on credit period for 30 days@8% p.a.	USD 13,333
Total Outflow in USD	USD 2,013,333
Applicable forward rate for 1 USD	₹ 57.50
Total Outflow in ₹ (USD 2,013,333 ×₹ 57.50)	₹ 115,766,648

Alternative 1 is better as it entails lower cash outflow.

Practical Q13

13. Followings are the spot exchange rates quoted at three different forex markets:

USD/INR	48.30 in Mumbai
GBP/INR	77.52 in London
GBP/USD	1.6231 in New York

The arbitrageur has USD1,00,00,000. Assuming that there are no transaction costs, explain whether there is any arbitrage gain possible from the quoted spot exchange rates.

Solution — ICAI

13. The arbitrageur can proceed as stated below to realize arbitrage gains.

(i) Buy ₹ from USD 10,000,000 At Mumbai 48.30 × 10,000,000 ₹483,000,000

(ii) Convert these ₹ to GBP at London $\left(\frac{₹\ 483,000,000}{₹\ 77.52}\right)$ GBP 6,230,650.155

(iii) Convert GBP to USD at New York GBP 6,230,650.155 × 1.6231 USD 10,112,968.26

There is net gain of USD 10,112968.26 less USD 10,000,000 i.e. USD 112,968.26

Practical Q14

14. The US dollar is selling in India at ₹ 55.50. If the interest rate for 6 months borrowing in India is 10% per annum and the corresponding rate in USA is 4%.

(i) Do you expect that US dollar will be at a premium or at discount in the Indian Forex Market?

(ii) What will be the expected 6-months forward rate for US dollar in India? and

(iii) What will be the rate of forward premium or discount?

Solution — ICAI

14. **(i)** Under the given circumstances, the USD is expected to quote at a premium in India as the interest rate is higher in India.

(ii) Calculation of the forward rate:

$$\frac{1+R_h}{1+R_f}=\frac{F_1}{E_o}$$

Where: R_h is home currency interest rate, R_f is foreign currency interest rate, F_1 is end of the period forward rate, and E_o is the spot rate.

Therefore $\frac{1+(0.10/2)}{1+(0.04/2)}=\frac{F_1}{55.50}$

$$\frac{1+0.05}{1+0.02}=\frac{F_1}{55.50}$$

or $\frac{1.05}{1.02}\times 55.50 = F_1$

or $\frac{58.275}{1.02} = F_1$

or F_1 = ₹57.13

(iii) Rate of premium:

$$\frac{57.13-55.50}{55.50}\times\frac{12}{6}\times 100 = 5.87\%$$

Practical QIS

15. In March, 2009, the Multinational Industries make the following assessment of dollar rates per British pound to prevail as on 1.9.2009:

$/Pound	Probability
1.60	0.15
1.70	0.20
1.80	0.25
1.90	0.20
2.00	0.20

(i) What is the expected spot rate for 1.9.2009?

(ii) If, as of March, 2009, the 6-month forward rate is $ 1.80, should the firm sell forward its pound receivables due in September, 2009?

Solution — ICAI

15. **(i)** Calculation of expected spot rate for September, 2009:

$ for £ *(1)*	*Probability* *(2)*	*Expected $/£* *(1) × (2) = (3)*
1.60	0.15	0.24
1.70	0.20	0.34
1.80	0.25	0.45
1.90	0.20	0.38
2.00	0.20	0.40
	1.00	EV = 1.81

Therefore, the expected spot value of $ for £ for September, 2009 would be $ 1.81.

(ii) If the six-month forward rate is $ 1.80, the expected profits of the firm can be maximised by retaining its pounds receivable.

Practical Q16

16. An importer customer of your bank wishes to book a forward contract with your bank on 3rd September for sale to him of SGD 5,00,000 to be delivered on 30th October.

The spot rates on 3rd September are USD 49.3700/3800 and USD/SGD 1.7058/68. The swap points are:

USD /₹		USD/SGD	
Spot/September	0300/0400	1st month forward	48/49
Spot/October	1100/1300	2nd month forward	96/97
Spot/November	1900/2200	3rd month forward	138/140
Spot/December	2700/3100		
Spot/January	3500/4000		

Calculate the rate to be quoted to the importer by assuming an exchange margin of 5 paisa.

Solution — ICAI

16.

USD/ ₹ on 3rd September	49.3800
Swap Point for October	0.1300
	49.5100
USD/ SGD on 3rd September	1.7058
Swap Point for 2nd month Forward	0.0096
	1.7154
SGD/ ₹ (49.5100/ 1.7154)	28.8621
Add: Exchange Margin	0.0500
	28.9121

Thus, Cross Rate for SGD/ ₹ of 30th October shall be ₹ 28.9121.

Practical Q17

17. A company operating in Japan has today affected sales to an Indian company, the payment being due 3 months from the date of invoice. The invoice amount is 108 lakhs yen. At today's spot rate, it is equivalent to ₹ 30 lakhs. It is anticipated that the exchange rate will decline by 10% over the 3 months period and in order to protect the yen payments, the importer proposes to take appropriate action in the foreign exchange market. The 3 months forward rate is presently quoted as 3.3 yen per rupee. You are required to calculate the expected loss and to show how it can be hedged by a forward contract.

Solution — ICAI

17. Spot rate of ₹ 1 against yen = 108 lakhs yen/₹ 30 lakhs = 3.6 yen

3 months forward rate of Re. 1 against yen = 3.3 yen

Anticipated decline in Exchange rate = 10%.

Expected spot rate after 3 months = 3.6 yen – 10% of 3.6 = 3.6 yen – 0.36 yen = 3.24 yen per rupee

	₹ (in lakhs)
Present cost of 108 lakhs yen	30.00
Cost after 3 months: 108 lakhs yen/ 3.24 yen	33.33
Expected exchange loss	3.33
If the expected exchange rate risk is hedged by a Forward contract:	
Present cost	30.00
Cost after 3 months if forward contract is taken 108 lakhs yen/ 3.3 yen	32.73
Expected loss	2.73

Suggestion: If the exchange rate risk is not covered with forward contract, the expected exchange loss is ₹ 3.33 lakhs. This could be reduced to ₹ 2.73 lakhs if it is covered with Forward contract. Hence, taking forward contract is suggested.

Practical Q18

18. ABC Co. have taken a 6 month loan from their foreign collaborators for US Dollars 2 millions. Interest payable on maturity is at SOFR plus 1.0%. Current 6-month SOFR is 2%.

Enquiries regarding exchange rates with their bank elicits the following information:

Spot USD 1	₹ 48.5275
6 months forward	₹ 48.4575

(i) What would be their total commitment in Rupees, if they enter into a forward contract?

(ii) Will you advise them to do so? Explain giving reasons.

Solution — ICAI

18. Firstly, the interest is calculated at 3% p.a. for 6 months. That is:

USD 20,00,000 × 3/100 × 6/12 = USD 30,000

From the forward points quoted, it is seen that the second figure is less than the first, this means that the currency is quoted at a discount.

(i) The value of the total commitment in Indian rupees is calculated as below:

Principal Amount of loan	USD 20,00,000
Add: Interest	USD 30,000
Amount due	USD 20,30,000
Spot rate	₹ 48.5275
Forward Points (6 months)	(–) 0.0700
Forward Rate	₹ 48.4575
Value of Commitment	₹ 9,83,68,725

(ii) It is seen from the forward rates that the market expectation is that the dollar will depreciate. If the firm's own expectation is that the dollar will depreciate more than what the bank has quoted, it may be worthwhile not to cover forward and keep the exposure open.

If the firm has no specific view regarding future dollar price movements, it would be better to cover the exposure. This would freeze the total commitment and insulate the firm from undue market fluctuations. In other words, it will be advisable to cut the losses at this point of time.

Given the interest rate differentials and inflation rates between India and USA, it would be unwise to expect continuous depreciation of the dollar. The US Dollar is a stronger currency than the Indian Rupee based on past trends and it would be advisable to cover the exposure.

Practical Q19

19. Excel Exporters are holding an Export bill in United States Dollar (USD) 1,00,000 due 60 days hence. They are worried about the falling USD value which is currently at ₹ 45.60 per USD. The concerned Export Consignment has been priced on an Exchange rate of ₹ 45.50 per USD. The Firm's Bankers have quoted a 60-day forward rate of ₹ 45.20.

Calculate:

(i) Rate of discount quoted by the Bank

(ii) The probable loss of operating profit if the forward sale is agreed to.

Solution — ICAI

19. (i) Rate of discount quoted by the bank

$$= \frac{(45.20 - 45.60) \times 365 \times 100}{45.60 \times 60} = 5.33\%$$

(ii) Probable loss of operating profit:

(45.20 – 45.50) × 1,00,000 = ₹ 30,000

Practical Q20

20. In International Monetary Market an international forward bid for December, 15 on pound sterling is $ 1.2816 at the same time that the price of IMM sterling future for delivery on December, 15 is $ 1.2806. The contract size of pound sterling is £ 62,500. How could the dealer use arbitrage in profit from this situation and how much profit is earned?

Solution — ICAI

20. Buy £ 62500 × 1.2806 = $ 80037.50

Sell £ 62500 × 1.2816 = $ 80100.00

Profit $ 62.50

Alternatively, if the market comes back together before December 15, the dealer could unwind his position (by simultaneously buying £ 62,500 forward and selling a futures contract. Both for delivery on December 15) and earn the same profit of $ 62.5.

Practical Q21

21. An Indian importer has to settle an import bill for $ 1,30,000. The exporter has given the Indian exporter two options:

(i) Pay immediately without any interest charges.

(ii) Pay after three months with interest at 5 percent per annum.

The importer's bank charges 15 percent per annum on overdrafts. The exchange rates in the market are as follows:

Spot rate (₹ /$) : 48.35 /48.36

3-Months forward rate (₹/$) : 48.81 /48.83

The importer seeks your advice. Give your advice.

Solution — ICAI

21. If importer pays now, he will have to buy US$ in Spot Market by availing overdraft facility. Accordingly, the outflow under this option will be

	₹
Amount required to purchase $130000[$130000 X ₹48.36]	6286800
Add: Overdraft Interest for 3 months @15% p.a.	235755
	6522555

If importer makes payment after 3 months then, he will have to pay interest for 3 months @ 5% p.a. for 3 month along with the sum of import bill. Accordingly, he will have to buy $ in forward market. The outflow under this option will be as follows:

	$
Amount of Bill	130000
Add: Interest for 3 months @5% p.a.	1625
	131625

Amount to be paid in Indian Rupee after 3 month under the forward purchase contract

₹ 6427249 (US$ 131625 X ₹ 48.83)

Since outflow of cash is least in (ii) option, it should be opted for.

Practical Q22

22. DEF Ltd. has imported goods to the extent of US$ 1 crore. The payment terms are 60 days interest-free credit. For additional credit of 30 days, interest at the rate of 7.75% p.a. will be charged.

The banker of DEF Ltd. has offered a 30 days loan at the rate of 9.5% p.a. Their quote for the foreign exchange is as follows:

Spot rate INR/US$	62.50
60 days forward rate INR/US$	63.15
90 days forward rate INR/US$	63.45

Which one of the following options would be better?

(i) Pay the supplier on 60th day and avail bank loan for 30 days.

(ii) Avail the supplier's offer of 90 days credit.

Solution — ICAI

22. (i) Pay the supplier in 60 days

If the payment is made to supplier in 60 days the applicable forward rate for 1 USD	₹ 63.15
Payment Due	USD 1 crore
Outflow in Rupees (USD 1 crore × ₹ 63.15)	₹ 63.15 crore
Add: Interest on loan for 30 days@9.5% p.a.	₹ 0.50 crore
Total Outflow in ₹	₹ 63.65 crore

(ii) Availing supplier's offer of 90 days credit

Amount Payable	USD 1.00000 crore
Add: Interest on credit period for 30 days@7.75% p.a.	USD 0.00646 crore
Total Outflow in USD	USD 1.00646 crore
Applicable forward rate for 1 USD	₹ 63.45
Total Outflow in ₹ (USD 1.00646 crore ×₹ 63.45)	₹ 63.86 crore

Alternative 1 is better as it entails lower cash outflow.

Practical Q23

23. A company is considering hedging its foreign exchange risk. It has made a purchase on 1st July, 2016 for which it has to make a payment of US$ 60,000 on December 31, 2016. The present exchange rate is 1 US $ = ₹ 65. It can purchase forward 1 $ at ₹ 64. The company will have to make an upfront premium @ 2% of the forward amount purchased. The cost of funds to the company is 12% per annum.

In the following situations, compute the profit/loss the company will make if it hedges its foreign exchange risk with the exchange rate on 31st December, 2016 as:

(i) ₹ 68 per US $.

(ii) ₹ 62 per US $.

(iii) ₹ 70 per US $.

(iv) ₹ 65 per US $.

Solution — ICAI

23.

	(₹)
Present Exchange Rate ₹65 = 1 US$	
If company purchases US$ 60,000 forward premium is	
60000 × 64 × 2%	76,800
Interest on ₹76,800 for 6 months at 12%	4,608
Total hedging cost	81,408
If exchange rate is ₹68	
Then gain (₹68 – ₹64) for US$ 60,000	2,40,000
Less: Hedging cost	81,408
Net gain	1,58,592
If US$ = ₹62	
Then loss (₹64 – ₹62) for US$ 60,000	1,20,000
Add: Hedging Cost	81,408
Total Loss	2,01,408
If US$ = ₹70	
Then Gain (₹70 – ₹64) for US$ 60,000	3,60,000
Less: Hedging Cost	81,408
Total Gain	2,78,592
If US$ = ₹65	
Then Gain (₹ 65 – ₹ 64) for US$ 60,000	60,000
Less: Hedging Cost	81,408
Net Loss	21,408

Practical Q24

24. Following information relates to AKC Ltd. which manufactures some parts of an electronics device which are exported to USA, Japan and Europe on 90 days credit terms.

Cost and Sales information:

	Japan	USA	Europe
Variable cost per unit	₹225	₹395	₹510
Export sale price per unit	Yen 650	US$10.23	Euro 11.99
Receipts from sale due in 90 days	Yen 78,00,000	US$1,02,300	Euro 95,920

Foreign exchange rate information:

	Yen/₹	US$/₹	Euro/₹
Spot market	2.417-2.437	0.0214-0.0217	0.0177-0.0180
3 months forward	2.397-2.427	0.0213-0.0216	0.0176-0.0178
3 months spot	2.423-2.459	0.02144-0.02156	0.0177-0.0179

Advise AKC Ltd. by calculating average contribution to sales ratio whether it should hedge its foreign currency risk or not.

Solution — ICAI

24. If foreign exchange risk is hedged

				Total (₹)
Sum due	Yen 78,00,000	US$1,02,300	Euro 95,920	
Unit input price	Yen 650	US$10.23	Euro 11.99	
Unit sold	12000	10000	8000	
Variable cost per unit	₹ 225/-	₹ 395/-	₹ 510/-	
Variable cost	₹ 27,00,000	₹ 39,50,000	₹ 40,80,000	₹ 1,07,30,000
Three months forward rate for selling	2.427	0.0216	0.0178	
Rupee value of receipts	₹32,13,844	₹ 47,36,111	₹ 53,88,764	₹ 1,33,38,719
Contribution	₹5,13,844	₹ 7,86,111	₹ 13,08,764	₹ 26,08,719
Average contribution to sale ratio				19.56%
If risk is not hedged				
Rupee value of receipt	₹31,72,021	₹ 47,44,898	₹ 53,58,659	₹ 1,32,75,578
Total contribution				₹ 25,45,578
Average contribution to sale ratio				19.17%

AKC Ltd. Is advised to hedge its foreign currency exchange risk.

Practical Q25

25. EFD Ltd. is an Indian export business house. The company prepares invoice in customers' currency. Its debtors of US$. 10,000,000 is due to be received on April 1, 2015.

Market information as at January 1, 2015 is:

Exchange rates US$/INR		Currency Futures US$/INR	
Spot	0.016667	Contract size: ₹ 24,816,975	
1-month forward	0.016529	1-month	0.016519
3-months forward	0.016129	3-month	0.016118

	Initial Margin	Interest rates in India
1-Month	₹ 17,500	6.5%
3-Months	₹ 22,500	7%

On April 1, 2015 the spot rate US$/INR is 0.016136 and Currency Futures rate is 0.016134.

Which of the following methods would be most advantageous to EFD Ltd?

(i) Using forward contract

(ii) Using currency futures

(iii) Not hedging the currency risk

Solution — ICAI

25. Receipts using a forward contract = $10,000,000/0.016129 = ₹ 620,001,240

Receipts using currency futures

The number of contracts needed is ($10,000,000/0.016118)/24,816,975 = 25

Initial margin payable is 25 contracts x ₹ 22,500 = ₹ 5,62,500

On April 1,2015 Futures Contracts close at 0.016134

Receipts = US$10,000,000/0.016136 = ₹ 619,732,276

Variation Margin =

[(0.016134 – 0.016118) x 25 x 24,816,975]/0.016136

OR (0.000016x 25 x 24,816,975)/.016136 = 9926.79/0.016136 = ₹ 615,195

Less: Interest Cost – ₹ 5,62,500x 0.07 x 3/12 = ₹ 9,844

Net Receipts ₹ 620,337,627

Receipts under different methods of hedging

Forward contract ₹ 620,001,240

Futures ₹ 620,337,627

No hedge (US$ 10,000,000/0.016136) ₹ 619,732,276

The most advantageous option would have been to hedge with Futures.

Practical Q26

26. Spot rate 1 US $ = ₹ 48.0123

180 days Forward rate for 1 US $ = ₹ 48.8190

Annualised interest rate for 6 months – Rupee = 12%

Annualised interest rate for 6 months – US $ = 8%

Is there any arbitrage possibility? If yes how an arbitrageur can take advantage of the situation, if he is willing to borrow ₹ 40,00,000 or US $83,312.

Solution — ICAI

26.

Spot Rate = ₹40,00,000 /US$83,312 = 48.0123

Forward Premium on US$ = [(48.8190 – 48.0123)/48.0123] x 12/6 x 100

= 3.36%

Interest rate differential = 12% - 8% = 4%

Since the negative Interest rate differential is greater than forward premium there is a possibility of arbitrage inflow into India.

The advantage of this situation can be taken in the following manner:

1. Borrow US$ 83,312 for 6 months

   Amount to be repaid after 6 months

   = US $ 83,312 (1+0.08 x 6/12) = US$86,644.48

2. Convert US$ 83,312 into Rupee and get the principal i.e. ₹40,00,000

   Interest on Investments for 6 months – ₹40,00,000/- x 0.06 = ₹2,40,000/-

   Total amount at the end of 6 months = ₹(40,00,000 + 2,40,000) = ₹42,40,000/-

   Converting the same at the forward rate

   = ₹42,40,000/ ₹48.8190= US$ 86,851.43

Hence the gain is US $ (86,851.43 – 86,644.48) = US$ 206.95 OR

₹10,103 i.e., ($206.95 x ₹48.8190)

Practical Q27

27. Given the following information:

Exchange rate – Canadian dollar 0.665 per DM (spot)

Canadian dollar 0.670 per DM (3 months)

Interest rates – DM 7% p.a.

Canadian Dollar – 9% p.a.

What operations would be carried out to take the possible arbitrage gains?

Solution — ICAI

27. In this case, DM is at a premium against the Can$.

Premium = [(0.67 – 0.665) /0.665] x (12/3) x 100 = 3.01 per cent

Interest rate differential = 9% - 7% = 2 per cent.

Since the interest rate differential is smaller than the premium, it will be profitable to place money in Deutschmarks the currency whose 3-months interest is lower.

The following operations are carried out:

(i) Borrow Can$ 1000 at 9 per cent for 3- months;

(ii) Change this sum into DM at the spot rate to obtain DM

= (1000/0.665) = 1503.76

(iii) Place DM 1503.76 in the money market for 3 months to obtain a sum of DM

Principal:	1503.76
Add: Interest @ 7% for 3 months =	26.32
Total	1530.08

(iv) Sell DM at 3-months forward to obtain Can$= (1530.08x0.67) = 1025.15

(v) Refund the debt taken in Can$ with the interest due on it, i.e.,

	Can$
Principal	1000.00
Add: Interest @9% for 3 months	22.50
Total	1022.50

Net arbitrage gain = 1025.15 – 1022.50 = Can$ 2.65

Practical Q28

28. An Indian exporting firm, Rohit and Bros., would be covering itself against a likely depreciation of pound sterling. The following data is given:

Receivables of Rohit and Bros	:	£500,000
Spot rate	:	₹ 56.00/£
Payment date	:	3-months
3 months interest rate	:	India : 12 per cent per annum
	:	UK : 5 per cent per annum

What should the exporter do?

Solution — ICAI

28. The only thing lefts Rohit and Bros to cover the risk in the money market. The following steps are required to be taken:

(i) Borrow pound sterling for 3- months. The borrowing has to be such that at the end of three months, the amount becomes £ 500,000. Say, the amount borrowed is £ x. Therefore

$$x\left[1+0.05\times\frac{3}{12}\right]=500{,}000 \text{ or } x = £493{,}827$$

(ii) Convert the borrowed sum into rupees at the spot rate. This gives: £493,827 × ₹ 56 = ₹ 27,654,312

(iii) The sum thus obtained is placed in the money market at 12 per cent to obtain at the end of 3- months:

$$S = ₹\ 27,654,312 \times \left[1 + 0.12 \times \frac{3}{12}\right] = ₹\ 28,483,941$$

(iv) The sum of £500,000 received from the client at the end of 3- months is used to refund the loan taken earlier.

From the calculations. It is clear that the money market operation has resulted into a net gain of ₹ 483,941 (₹ 28,483,941 – ₹ 500,000 × 56).

If pound sterling has depreciated in the meantime. The gain would be even bigger.

Practical Q29

29. An exporter is a UK based company. Invoice amount is $3,50,000. Credit period is three months. Exchange rates in London are :

Spot Rate ($/£) 1.5865 – 1.5905

3-month Forward Rate ($/£) 1.6100 – 1.6140

Rates of interest in Money Market :

	Deposit	Loan
$	7%	9%
£	5%	8%

Compute and show how a money market hedge can be put in place. Compare and contrast the outcome with a forward contract.

Solution — ICAI

29. Identify: Foreign currency is an asset. Amount $ 3,50,000.

Create: $ Liability.

Borrow: In $. The borrowing rate is 9% per annum or 2.25% per quarter.

Amount to be borrowed: 3,50,000 / 1.0225 = $ 3,42,298.29

Convert: Sell $ and buy £. The relevant rate is the Ask rate, namely, 1.5905 per £,

(**Note:** This is an indirect quote). Amount of £s received on conversion is 2,15,214.27 (3,42,298.29/1.5905).

Invest: £ 2,15,214.27 will be invested at 5% for 3 months and get £ 2,17,904.45

Settle: The liability of $3,42,298.29 at interest of 2.25 per cent quarter matures to $3,50,000 receivable from customer.

Using forward rate, amount receivable is = 3,50,000 / 1.6140 = £2,16,852.54

Amount received through money market hedge = £2,17,904.45

Gain = 2,17,904.45 – 2,16,852.54 = £1,051.91

So, money market hedge is beneficial for the exporter.

Practical Q30

30. The rate of inflation in India is 8% per annum and in the U.S.A. it is 4%. The current spot rate for USD in India is ₹ 46. What will be the expected rate after 1 year and after 4 years applying the Purchasing Power Parity Theory.

Solution — ICAI

30.

End of Year	₹	₹/USD
1	$₹46.00 \times \frac{(1+0.08)}{(1+0.04)}$	47.77
2	$₹47.77 \times \frac{(1+0.08)}{(1+0.04)}$	49.61
3	$₹49.61 \times \frac{(1+0.08)}{(1+0.04)}$	51.52
4	$₹51.52 \times \frac{(1+0.08)}{(1+0.04)}$	53.50

Practical Q31

31. On April 1, 3 months interest rate in the UK £ and US $ are 7.5% and 3.5% per annum respectively. The UK £/US $ spot rate is 0.7570. What would be the forward rate for US $ for delivery on 30th June?

Solution — ICAI

31. As per interest rate parity

$$S_1 = S_0\left[\frac{1+\text{in A}}{1+\text{in B}}\right]$$

$$S_1 = £0.7570\left[\frac{1+(0.075)\times 3/12}{1+(0.035)\times 3/12}\right]$$

$$= £0.7570\left[\frac{1.01875}{1.00875}\right]$$

$$= £0.7570 \times 1.0099 = £0.7645$$

= UK £0.7645 / US$

Practical Q32

32. An importer requests his bank to extend the forward contract for US$ 20,000 which is due for maturity on 30th October, 2010, for a further period of 3 months. He agrees to pay the required margin money for such extension of the contract.

Contracted Rate – US$ 1= ₹ 42.32

The US Dollar quoted on 30-10-2010:-

Spot – 41.5000/41.5200

3 months' Premium - 0.87% /0.93%

Margin money for buying and selling rate is 0.075% and 0.20% respectively.

Compute:

(i) The cost to the importer in respect of the extension of the forward contract, and

(ii) The rate of new forward contract.

Solution — ICAI

32. (i) The contract is to be cancelled on 30-10-2010 at the spot buying rate of US$ 1 = ₹ 41.5000

Less: Margin Money 0.075%	= ₹ 0.0311
	= ₹ 41.4689 or ₹ 41.47
US$ 20,000 @ ₹ 41.47	= ₹ 8,29,400
US$ 20,000 @ ₹ 42.32	= ₹ 8,46,400
The difference in favour of the Bank/Cost to the importer	₹ 17,000

(ii) The Rate of New Forward Contract

Spot Selling Rate US$ 1	= ₹ 41.5200
Add: Premium @ 0.93%	= ₹ 0.3861
	= ₹ 41.9061
Add: Margin Money 0.20%	= ₹ 0.0838
	= ₹ 41.9899 or ₹ 41.99

Practical Q33

33. XYZ Ltd., an Indian firm, will need to pay JAPANESE YEN (JY) 5,00,000 on 30th June. In order to hedge the risk involved in foreign currency transaction, the firm is considering two alternative methods i.e. forward market cover and currency option contract.

On 1st April, following quotations (JY/INR) are made available:

Spot	3 months forward
1.9516/1.9711.	1.9726./1.9923

The prices for forex currency option on purchase are as follows:

Strike Price	JY 2.125
Call option (June)	JY 0.047
Put option (June)	JY 0.098

For excess or balance of JY covered, the firm would use forward rate as future spot rate.

You are required to recommend cheaper hedging alternative for XYZ LTD.

Solution — ICAI

33. (i) Forward Cover

3-month Forward Rate = $\frac{1}{1.9726}$ = ₹ 0.5070/JY

Accordingly, INR required for JY 5,00,000 (5,00,000 X ₹ 0.5070) ₹ 2,53,500

(ii) Option Cover

To purchase JY 5,00,000, XYZ LTD shall enter into a Put Option @ JY 2.125/INR

Accordingly, outflow in INR $\left(\frac{\text{JY } 5,00,000}{2.125}\right)$	₹ 2,35,294
Premium $\left(\frac{\text{INR } 2,35,294 \times 0.098}{1.9516}\right)$	₹ 11,815
	₹ 2,47,109

Since outflow of cash is least in case of Option same should be opted for. Further if price of INR goes above JY 2.125/INR the outflow shall further be reduced.

Practical Q34

34. ABC Technologic is expecting to receive a sum of US$ 4,00,000 after 3 months. The company decided to go for future contract to hedge against the risk. The standard size of future contract available in the market is $1000. As on date spot and futures $ contract are quoting at ₹ 44.00 & ₹ 45.00 respectively. Suppose after 3 months the company closes out its position futures are quoting at ₹ 44.50 and spot rate is also quoting at ₹ 44.50. You are required to calculate effective realization for the company while selling the receivable. Also calculate how company has been benefitted by using the future option.

Solution — ICAI

34. The company can hedge position by selling future contracts as it will receive amount from outside.

Number of Contracts = $\frac{\$4,00,000}{\$1,000}$ = 400 contracts

Gain by trading in futures = (₹ 45 – ₹ 44.50) 4,00,000= ₹ 2,00,000

Net Inflow after after 3 months = ₹ 44.50 x ₹ 4,00,000+ 2,00,000 = ₹ 1,80,00,000

Effective Price realization = $\frac{₹\ 1,80,00,000}{\$\ 4,00,000}$ = ₹ 45 Per US$

Practical Q35

35. Gibralater Limited has imported 5000 bottles of shampoo at landed cost in Mumbai, of US $ 20 each. The company has the choice for paying for the goods immediately or in 3 months' time. It has a clean overdraft limit where 14% p.a. rate of interest is charged.

Calculate which of the following method would be cheaper to Gibralter Limited.

(i) Pay in 3 months' time with interest @ 10% p.a. and cover risk forward for 3 months.

(ii) Settle now at a current spot rate and pay interest of the overdraft for 3 months.

The rates are as follows:

Mumbai ₹ /$ spot	:	60.25-60.55
3 months swap points	:	35/25

Solution — ICAI

35. Option - I

$20 x 5000 = $ 1,00,000

Repayment in 3 months time = $1,00,000 x (1 + 0.10/4) = $ 1,02,500

3-months outright forward rate = ₹ 59.90/ ₹ 60.30

Repayment obligation in ₹ ($1,02,500 X ₹ 60.30) = ₹ 61,80,750

Option - II

Overdraft ($1,00,000 x ₹ 60.55)	₹ 60,55,000
Interest on Overdraft (₹ 60,55,000 x 0.14/4)	₹ 2,11,925
	₹ 62,66,925

Option I should be preferred as it has lower outflow.

Practical Q36

36. An American firm is under obligation to pay interests of Can$ 1010000 and Can$ 705000 on 31st July and 30th September respectively. The Firm is risk averse and its policy is to hedge the risks involved in all foreign currency transactions. The Finance Manager of the firm is thinking of hedging the risk considering two methods i.e. fixed forward or option contracts.

It is now June 30. Following quotations regarding rates of exchange, US$ per Can$, from the firm's bank were obtained:

Spot	1 Month Forward	3 Months Forward
0.9284-0.9288	0.9501	0.9556

Price for a Can$ option on a U.S. stock exchange (cents per Can$, payable on purchase of the option, contract size Can$ 50000) are as follows:

Strike Price	Calls		Puts	
(US$/Can$)	July	Sept.	July	Sept.
0.93	1.56	2.56	0.88	1.75
0.94	1.02	NA	NA	NA
0.95	0.65	1.64	1.92	2.34

According to the suggestion of finance manager if options are to be used, one month option should be bought at a strike price of 94 cents and three month option at a strike price of 95 cents and for the remainder uncovered by the options the firm would bear the risk itself. For this, it would use forward rate as the best estimate of spot. Transaction costs are ignored.

Recommend, which of the above two methods would be appropriate for the American firm to hedge its foreign exchange risk on the two interest payments.

Solution — ICAI

36. *Forward Market Cover*

Hedge the risk by buying Can$ in 1 and 3 months time will be:

July - 1010000 X 0.9501 = US $ 959601

Sept. - 705000 X 0.9556 = US $ 673698

Option Contracts

July Payment = 1010000/ 50,000 = 20.20

September Payment = 705000/ 50,000 = 14.10

Company would like to take out 20 contracts for July and 14 contracts for September respectively. Therefore costs, if the options were exercised, will be:

	July		Sept.	
	Can $	US $	Can $	US $
Covered by Contracts	1000000	940000	700000	665000
Balance bought at spot rate	10000	9501	5000	4778
Option Costs:				
Can $ 50000 x 20 x 0.0102		10200	---	
Can $ 50000 x 14 x 0.0164	---			11480
Total cost in US $ of using Option Contract		959701		681258

Decision: As the firm is stated as risk averse and the money due to be paid is certain, a fixed forward contract, being the cheapest alternative in the both the cases, would be recommended.

Practical Q37

37. Zaz plc, a UK Company is in the process of negotiating an order amounting €2.8 million with a large German retailer on 6 month's credit. If successful, this will be first time for Zaz has exported goods into the highly competitive German Market. The Zaz is considering following 3 alternatives for managing the transaction risk before the order is finalized.

(a) Mr. Peter the Marketing head has suggested that in order to remove transaction risk completely Zaz should invoice the German firm in Sterling using the current €/£ average spot rate to calculate the invoice amount.

(b) Mr. Wilson, CE is doubtful about Mr. Peter's proposal and suggested an alternative of invoicing the German firm in € and using a forward exchange contract to hedge the transaction risk.

(c) Ms. Karen, CFO is agreed with the proposal of Mr. Wilson to invoice the German first in €, but she is of opinion that Zaz should use sufficient 6-month Sterling Future contracts (to the nearest whole number) to hedge the transaction risk.

Following data is available

Spot Rate	€ 1.1960 - €1.1970/£
6-months forward points	0.60 – 0.55 Euro Cents.
6-month Future contract is currently trading at	€ 1.1943/£
6-month Future contract size is	£62,500
After 6-month Spot rate and future rate	€ 1.1873/£

You are required to

(a) Calculate (to the nearest £) the £ receipt for Zaz plc, under each of 3 above proposals.

(b) In your opinion which alternative you consider to be most appropriate.

Solution — ICAI

37. (i) Receipt under three proposals

(a) Proposal of Mr. Peter

Invoicing in £ will produce = $\frac{€\,2.8\text{ million}}{1.1965}$ = £ 2.340 million

(b) Proposal of Mr. Wilson

Forward Rate = € 1.1970-0.0055 = 1.1915

Using Forward Market hedge Sterling receipt would be $\frac{€\,2.8\text{million}}{1.1915}$ = £ 2.35 million

(c) Proposal of Ms. Karen

The equivalent sterling of the order placed based on future price (€1.1943)

= $\frac{€\,2.8\text{million}}{1.1943}$ = £ 2,344,470 (rounded off)

Number of Contracts = $\frac{£2{,}344{,}470}{62{,}500}$ = 37 Contracts (to the nearest whole number)

Thus, € amount hedged by future contract will be = 37 × £ 62,500 = £ 23,12,500

Buy Future at	€ 1.1943
Sell Future at	€ 1.1873
	€ 0.0070

Total loss on Future Contracts = 37 × £ 62,500 × € 0.0070 = € 16,188

After 6 months

Amount Received	€ 28,00,000
Less: Loss on Future Contracts	€ 16,188
	€ 27,83,812

Sterling Receipts

On sale of € at spot = $\frac{€\,27{,}83{,}812}{1.1873}$ = £ 2.3446 million

(ii) Proposal of option (b) is preferable because the option (a) & (c) produces least receipts. Further, in case of proposal (a) there must be a doubt as to whether this would be acceptable to German firm as it is described as a competitive market and Zaz is moving into it first time.

Practical Q38

38. Columbus Surgicals Inc. is based in US, has recently imported surgical raw materials from the UK and has been invoiced for £ 480,000, payable in 3 months. It has also exported surgical goods to India and France.

The Indian customer has been invoiced for £ 138,000, payable in 3 months, and the French customer has been invoiced for € 590,000, payable in 4 months.

Current spot and forward rates are as follows:

£ / US$

Spot:	0.9830 – 0.9850
Three months forward:	0.9520 – 0.9545

US$ / €

Spot:	1.8890 – 1.8920
Four months forward:	1.9510 – 1.9540

Current money market rates are as follows:

UK:	10.0% – 12.0% p.a.
France:	14.0% – 16.0% p.a.
USA:	11.5% – 13.0% p.a.

You as Treasury Manager are required to show how the company can hedge its foreign exchange exposure using Forward markets and Money markets hedge and suggest which the best hedging technique is.

Solution — ICAI

38. £ Exposure

Since Columbus has a £ receipt (£ 138,000) and payment of (£ 480,000) maturing at the same time i.e. 3 months, it can match them against each other leaving a net liability of £ 342,000 to be hedged.

(i) Forward market hedge

Buy 3 months' forward contract accordingly, amount payable after 3 months will be

£ 342,000 / 0.9520 = US$ 359,244

(ii) Money market hedge

To pay £ after 3 months' Columbus shall requires to borrow in US$ and translate to £ and then deposit in £.

For payment of £ 342,000 in 3 months (@2.5% interest) amount required to be deposited now (£ 342,000 ÷ 1.025) = £ 333,658

With spot rate of 0.9830 the US$ loan needed will be = US$ 339,429

Loan repayable after 3 months (@3.25% interest) will be = US$ 350,460

In this case the money market hedge is a cheaper option.

€ Receipt

Amount to be hedged = € 590,000

(i) Forward market hedge

Sell 4 months' forward contract accordingly, amount receivable after 4 months will be (€ 590,000 x1.9510) = US$ 1,151,090

(ii) Money market hedge

For money market hedge Columbus shall borrow in € and then translate to US$ and deposit in US$

For receipt of € 590,000 in 4 months (@ 5.33% interest) amount required to be borrowed now (€590,000 ÷ 1.0533) = € 560,144

With spot rate of 1.8890 the US$ deposit will be = US$ 1,058,113

Deposit amount will increase over 4 months (@3.83% interest) will be = US$ 1,098,639

In this case, more will be received in US$ under the forward hedge.

Practical Q39

39. XYZ Ltd. a US firm will need £ 3,00,000 in 180 days. In this connection, the following information is available:

Spot rate 1 £ = $ 2.00

180 days forward rate of £ as of today = $1.96

Interest rates are as follows:

	U.K.	US
180 days deposit rate	4.5%	5%
180 days borrowing rate	5%	5.5%

A call option on £ that expires in 180 days has an exercise price of $ 1.97 and a premium of $ 0.04.

XYZ Ltd. has forecasted the spot rates 180 days hence as below:

Future rate	Probability
$ 1.91	25%
$ 1.95	60%
$ 2.05	15%

Which of the following strategies would be most preferable to XYZ Ltd.?

(a) A forward contract;

(b) A money market hedge;

(c) An option contract;

(d) No hedging.

Show calculations in each case

Solution — ICAI

39. (a) Forward contract: Dollar needed in 180 days = £3,00,000 x $ 1.96 = $5,88,000/-

(b) Money market hedge: Borrow $, convert to £, invest £, repay $ loan in 180 days

Amount in £ to be invested = 3,00,000/1.045 = £ 2,87,081

Amount of $ needed to convert into £ = 2,87,081 x 2 = $ 5,74,162

Interest and principal on $ loan after 180 days = $5,74,162 x 1.055 = $ 6,05,741

(c) Call option:

Expected Spot rate in 180 days	*Prem./ unit*	*Exercise Option*	*Total price per unit*	*Total price for £3,00,000 x i*	*Prob. Pi*	*pixi*
1.91	0.04	No	1.95	5,85,000	0.25	1,46,250
1.95	0.04	No	1.99	5,97,000	0.60	3,58,200
2.05	0.04	Yes	2.01*	6,03,000	0.15	90,450
						5,94,900
Add: Interest on Premium @ 5.5% (12,000 x 5.5%)						660
						5,95,560

* ($1.97 + $0.04)

(d) No hedge option:

Expected Future spot rate	*Dollar needed Xi*	*Prob. Pi*	*Pi xi*
1.91	5,73,000	0.25	1,43,250
1.95	5,85,000	0.60	3,51,000
2.05	6,15,000	0.15	92,250
			5,86,500

The probability distribution of outcomes for no hedge strategy appears to be most preferable because least number of $ are needed under this option to arrange £3,00,000.

Practical Q40

40. A Ltd. of U.K. has imported some chemical worth of USD 3,64,897 from one of the U.S. suppliers. The amount is payable in six months time. The relevant spot and forward rates are:

Spot rate	USD 1.5617-1.5673
6 months' forward rate	USD 1.5455 –1.5609

The borrowing rates in U.K. and U.S. are 7% and 6% respectively and the deposit rates are 5.5% and 4.5% respectively.

Currency options are available under which one option contract is for GBP 12,500. The option premium for GBP at a strike price of USD 1.70/GBP is USD 0.037 (call option) and USD 0.096 (put option) for 6 months period.

The company has 3 choices:

(i) Forward cover

(ii) Money market cover, and

(iii) Currency option

Which of the alternatives is preferable by the company?

Solution — ICAI

40. In the given case, the exchange rates are indirect. These can be converted into direct rates as follows:

Spot rate

GBP = $\frac{1}{\text{USD}1.5617}$ to $\frac{1}{\text{USD}1.5673}$

USD = GBP 0.64033 - GBP 0.63804

6 months' forward rate

GBP = $\frac{1}{\text{USD}1.5455}$ to $\frac{1}{\text{USD}1.5609}$

USD = GBP 0.64704 - GBP 0.64066

Payoff in 3 alternatives

i. Forward Cover

Amount payable	USD 3,64,897
Forward rate	GBP 0.64704
Payable in GBP	GBP 2,36,103

ii. Money market Cover

Amount payable	USD 3,64,897
PV @ 4.5% for 6 months i.e. $\frac{1}{1.0225} = 0.9779951$	USD 3,56,867
Spot rate purchase	GBP 0.64033
Borrow GBP 3,56,867 x 0.64033	GBP 2,28,512
Interest for 6 months @ 7 %	7,998
Payable after 6 months	GBP 2,36,510

iii. Currency options

Amount payable	USD 3,64,897
Unit in Options contract	GBP 12,500
Value in USD at strike rate of 1.70 (GBP 12,500 x 1.70)	USD 21,250
Number of contracts USD 3,64,897/ USD 21,250	17.17
Exposure covered USD 21,250 x 17	USD 3,61,250
Exposure to be covered by Forward (USD 3,64,897 – USD 3,61,250)	USD 3,647
Options premium 17 x GBP 12,500 x 0.096	USD 20,400
Premium in GBP (USD 20,400 x 0.64033)	GBP 13,063
Total payment in currency option	
Payment under option (17 x 12,500)	GBP 2,12,500
Premium payable	GBP 13,063
Payment for forward cover (USD 3,647 x 0.64704)	GBP 2,360
	GBP 2,27,923

Thus total payment in:

(i)	Forward Cover	2,36,103 GBP
(ii)	Money Market	2,36,510 GBP
(iii)	Currency Option	2,27,923 GBP

The company should take currency option for hedging the risk.

Note: Even interest on Option Premium can also be considered in the above solution.

Practical Q41

41. Nitrogen Ltd, a UK company is in the process of negotiating an order amounting to €4 million with a large German retailer on 6 months credit. If successful, this will be the first time that Nitrogen Ltd has exported goods into the highly competitive German market. The following three alternatives are being considered for managing the transaction risk before the order is finalized.

(i) Invoice the German firm in Sterling using the current exchange rate to calculate the invoice amount.

(ii) Alternative of invoicing the German firm in € and using a forward foreign exchange contract to hedge the transaction risk.

(iii) Invoice the German first in € and use sufficient 6 months sterling future contracts (to the nearly whole number) to hedge the transaction risk.

Following data is available:

Spot Rate	€ 1.1750 - €1.1770/£
6 months forward premium	0.55-0.60 Euro Cents
6 months future contract is currently trading at	€ 1.1760/£
6 months future contract size is	£ 62500
Spot rate and 6 months future rate	€ 1.1785/£

Required:

(a) Calculate to the nearest £ the receipt for Nitrogen Ltd, under each of the three proposals.

(b) In your opinion, which alternative would you consider to be the most appropriate and the reason thereof.

Solution — ICAI

41. (i) Receipt under three proposals

(a) Invoicing in Sterling

Invoicing in £ will produce $= \frac{€4\,\text{million}}{1.1770} = £3398471$

(b) Use of Forward Contract

Forward Rate = €1.1770 + 0.0060 = 1.1830

Using Forward Market hedge Sterling receipt would be $\frac{€4\text{ million}}{1.1830}$

= £ 3381234

(c) Use of Future Contract

The equivalent sterling of the order placed based on future price (€1.1760)

$= \frac{€4.00\,\text{million}}{1.1760} = £\ 3401360$

Number of Contracts = $\frac{£3401360}{62{,}500}$ = 54 Contracts (to the nearest whole number)

Thus, € amount hedged by future contract will be = 54 × £62,500 = £3375000

Buy Future at	€1.1760
Sell Future at	€1.1785
	€0.0025

Total profit on Future Contracts = 54 × £62,500 × €0.0025 = €8438

After 6 months

Amount Received	€ 4000000
Add: Profit on Future Contracts	€ 8438
	€ 4008438

Sterling Receipts

On sale of € at spot = $\frac{€4008438}{1.1785}$ = €3401305

(ii) Proposal of option (c) is preferable because the option (a) & (b) produces least receipts.

Practical Q42

42. Sun Ltd. is planning to import equipment from Japan at a cost of 3,400 lakh yen. The company may avail loans at 18 percent per annum with quarterly rests with which it can import the equipment. The company has also an offer from Osaka branch of an India based bank extending credit of 180 days at 2 percent per annum against opening of an irrecoverable letter of credit.

Additional information:

Present exchange rate	₹ 100 = 340 yen
180 day's forward rate	₹ 100 = 345 yen

Commission charges for letter of credit at 2 per cent per 12 months.

Advice the company whether the offer from the foreign branch should be accepted.

Solution — ICAI

42. **Option I (To finance the purchases by availing loan at 18% per annum):**

Cost of equipment	₹ in lakhs
3400 lakh yen at ₹100 = 340 yen	1,000.00
Add: Interest at 4.5% I Quarter	45.00
Add: Interest at 4.5% II Quarter (on ₹1045 lakhs)	47.03
Total outflow in Rupees	1,092.03

Alternatively, interest may also be calculated on compounded basis, i.e., ₹1000 × [1.045]²	₹1092.03

Option II (To accept the offer from foreign branch):

Cost of letter of credit	
At 1 % on 3400 lakhs yen at ₹100 = 340 yen	₹ 10.00 lakhs
Add: Interest for 2 Quarters	₹ 0.90 lakhs
(A)	₹ 10.90 lakhs
Payment at the end of 180 days:	
Cost	3400.00 lakhs yen
Interest at 2% p.a. [3400 × 2/100 × 180/365]	33.53 lakhs yen
	3433.53 lakhs yen
Conversion at ₹100 = 345 yen [3433.53 / 345 ×100] (B)	₹ 995.23 lakhs
Total Cost: (A) + (B)	₹ 1006.13 lakhs

Advise: Option 2 is cheaper by (1092.03 – 1006.13) lakh or ₹ 85.90 lakh. Hence, the offer may be accepted.

Practical Q43

43. NP and Co. has imported goods for US $ 7,00,000. The amount is payable after three months. The company has also exported goods for US $ 4,50,000 and this amount is receivable in two months. For receivable amount a forward contract is already taken at ₹ 48.90.

The market rates for Rupee and Dollar are as under:

Spot	₹ 48.50/70
Two months	25/30 points
Three months	40/45 points

The company wants to cover the risk and it has two options as under:

(A) To cover payables in the forward market and

(B) To lag the receivables by one month and cover the risk only for the net amount. No interest for delaying the receivables is earned.

Evaluate both the options if the cost of Rupee Funds is 12%. Which option is preferable?

Solution — ICAI

43. (A) To cover payable and receivable in forward Market

Amount payable after 3 months	$7,00,000
Forward Rate	₹ 48.45
Thus Payable Amount (₹) (A)	₹ 3,39,15,000
Amount receivable after 2 months	$ 4,50,000
Forward Rate	₹ 48.90
Thus Receivable Amount (₹) (B)	₹ 2,20,05,000
Interest @ 12% p.a. for 1 month (C)	₹ 2,20,050
Net Amount Payable in (₹) (A) – (B) – (C)	₹ 1,16,89,950

(B) Assuming that since the forward contract for receivable was already booked it shall be cancelled if we lag the receivables. Accordingly, any profit/ loss on cancellation of contract shall also be calculated and shall be adjusted as follows:

Amount Payable ($)	$7,00,000
Amount receivable after 3 months	$ 4,50,000
Net Amount payable	$2,50,000
Applicable Rate	₹ 48.45
Amount payable in (₹) (A)	₹ 1,21,12,500
Profit on cancellation of Forward cost (48.90 – 48.30) × 4,50,000 (B)	₹ 2,70,000
Thus, net amount payable in (₹) (A) + (B)	₹ 1,18,42,500

Since net payable amount is least in case of first option, hence the company should cover payable and receivables in forward market.

Note: 1. In the question it has not been clearly mentioned that whether quotes given for 2 and 3 months (in points terms) are premium points or direct quotes. Although above solution is based on the assumption that these are direct quotes, but students can also consider them as premium points and solve the question accordingly.

2. Alternatively, students can also consider the interest element on the Profit earned due cancellation of Forward Contract.

Practical Q44

44. A customer with whom the Bank had entered into 3 months' forward purchase contract for Swiss Francs 10,000 at the rate of ₹ 27.25 comes to the bank after 2 months and requests cancellation of the contract. On this date, the rates, prevailing, are:

Spot	CHF 1 =	₹ 27.30	27.35
One month forward		₹ 27.45	27.52

What is the loss/gain to the customer on cancellation?

Solution — ICAI

44. The contract would be cancelled at the one-month forward sale rate of ₹ 27.52.

	₹
Francs bought from customer under original forward contract at:	27.25
It is sold to him on cancellation at:	27.52
Net amount payable by customer per Franc	0.27

At ₹ 0.27 per Franc, exchange difference for CHF 10,000 is ₹ 2,700.

Loss to the Customer:

Exchange difference (Loss) ₹ 2,700

Note: The exchange commission and other service charges are ignored.

Practical Q45

45. A bank enters into a forward purchase TT covering an export bill for Swiss Francs 1,00,000 at ₹ 32.4000 due 25th April and covered itself for same delivery in the local interbank market at ₹ 32.4200. However, on 25th March, exporter sought for cancellation of the contract as the tenor of the bill is changed.

In Singapore market, Swiss Francs were quoted against dollars as under:

Spot	USD 1 = Sw. Fcs.	1.5076/1.5120
One month forward		1.5150/ 1.5160
Two months forward		1.5250 / 1.5270
Three months forward		1.5415/ 1.5445

and in the interbank market US dollars were quoted as under:

Spot	USD 1 = ₹	49.4302/4455
Spot / April		4100/4200
Spot/May		4300/4400
Spot/June		4500/4600

Calculate the cancellation charges, payable by the customer if exchange margin required by the bank is 0.10% on buying and selling.

Solution — ICAI

45. First the contract will be cancelled at TT Selling Rate

USD/ Rupee Spot Selling Rate	₹ 49.4455	
Add: Premium for April	₹ 0.4200	
	₹ 49.8655	
Add: Exchange Margin @ 0.10%	₹ 0.04987	
	₹ 49.91537	Or 49.9154
USD/ Sw. Fcs One Month Buying Rate	Sw. Fcs. 1.5150	
Sw. Fcs. Spot Selling Rate (₹49.91537/1.5150)	₹ 32.9474	
Rounded Off	₹ 32.9475	
Bank buys Sw. Fcs. Under original contract	₹ 32.4000	
Bank Sells under Cancellation	₹ 32.9475	
Difference payable by customer	₹ 00.5475	

Exchange difference of Sw. Fcs. 1,00,000 payable by customer ₹ 54,750

(Sw. Fcs. 1,00,000 x ₹ 0.5475)

Practical Q46

46. Your forex dealer had entered into a cross currency deal and had sold US $ 10,00,000 against EURO at US $ 1 = EURO 1.4400 for spot delivery.

However, later during the day, the market became volatile and the dealer in compliance with his management's guidelines had to square – up the position when the quotations were:

Spot US $ 1	INR 31.4300/4500
1 month margin	25/20
2 months margin	45/35
Spot US $ 1	EURO 1.4400/4450
1 month forward	1.4425/4490
2 months forward	1.4460/4530

What will be the gain or loss in the transaction?

Solution — ICAI

46. The amount of EURO bought by selling US$

US$ 10,00,000 * EURO 1.4400	=	EURO 14,40,000
The amount of EURO sold for buying USD 10,00,000 * 1.4450	=	EURO 14,45,000
Net Loss in the Transaction	=	EURO 5,000

To acquire EURO 5,000 from the market @

(a) USD 1 = EURO 1.4400 &

(b) USD1 = INR 31.4500

Cross Currency buying rate of EUR/INR is ₹ 31.4500 / 1.440 i.e. ₹ 21.8403

Loss in the Transaction ₹ 21.8403 * 5000 = ₹ 1,09,201.50

Alternatively, if delivery to be affected then computation of loss shall be as follows:

EURO to be surrendered to acquire $ 10,00,000	=	EURO 14,45,000
EURO to be received after selling $ 10,00,000	=	EURO 14,40,000
Loss	=	EURO 5,000

To acquire EURO 5,000 from market @

US $ 1 = EURO 1.4400

US $ 1 = INR 31.45

Cross Currency = $\frac{31.45}{1.440}$ = ₹ 21.8403

Loss in Transaction (21.8403 x EURO 5,000) = ₹ 1,09,201.50

Practical Q47

47. You have following quotes from Bank A and Bank B:

	Bank A	Bank B
SPOT	USD/CHF 1.4650/55	USD/CHF 1.4653/60
3 months	5/10	
6 months	10/15	
SPOT	GBP/USD 1.7645/60	GBP/USD 1.7640/50
3 months	25/20	
6 months	35/25	

Calculate :

(i) How much minimum CHF amount you have to pay for 1 Million GBP spot?

(ii) Considering the quotes from Bank A only, for GBP/CHF what are the Implied Swap points for Spot over 3 months?

Solution — ICAI

47. (i) To Buy 1 Million GBP Spot against CHF

1. First to Buy USD against CHF at the cheaper rate i.e. from Bank A. 1 USD = CHF 1.4655
2. Then to Buy GBP against USD at a cheaper rate i.e. from Bank B 1 GBP = USD 1.7650

By applying chain rule Buying rate would be

1 GBP = 1.7650 * 1.4655 CHF

1 GBP = CHF 2.5866

Amount payable CHF 2.5866 Million or CHF 25,86,600

(ii) Spot rate Bid rate GBP 1 = CHF 1.4650 * 1.7645 = CHF 2.5850

Offer rate GBP 1 = CHF 1.4655 * 1.7660 = CHF 2.5881

GBP / USD 3 months swap points are at discount

Outright 3 Months forward rate GBP 1 = USD 1.7620 / 1.7640

USD / CHF 3 months swap points are at premium

Outright 3 Months forward rate USD 1 = CHF 1.4655 / 1.4665

Hence

Outright 3 Months forward rate GBP 1 = CHF 2.5822 / 2.5869

Spot rate GBP 1 = CHF 2.5850 / 2.5881

Therefore 3-month swap points are at discount of 28/12.

Practical Q48

48. M/s Omega Electronics Ltd. exports air conditioners to Germany by importing all the components from Singapore. The company is exporting 2,400 units at a price of Euro 500 per unit. The cost of imported components is S$ 800 per unit. The fixed cost and other variables cost per unit are ₹ 1,000 and ₹ 1,500 respectively. The cash flows in Foreign currencies are due in six months. The current exchange rates are as follows:

₹/Euro	51.50/55
₹/S$	27.20/25

After six months the exchange rates turn out as follows:

₹/Euro	52.00/05
₹/S$	27.70/75

(A) You are required to calculate loss/gain due to transaction exposure.

(B) Based on the following additional information calculate the loss/gain due to transaction and operating exposure if the contracted price of air conditioners is ₹ 25,000 :

(i) the current exchange rate changes to

₹/Euro	51.75/80
₹/S$	27.10/15

(ii) Price elasticity of demand is estimated to be 1.5

(iii) Payments and receipts are to be settled at the end of six months.

Solution — ICAI

48. (i) Profit at current exchange rates

2400 [€ 500 × ₹ 51.50 – (S$ 800 × ₹ 27.25 + ₹ 1,000 + ₹ 1,500)]

2400 [₹ 25,750 - ₹ 24,300] = ₹ 34,80,000

Profit after change in exchange rates

2400[€500× ₹ 52 – (S$ 800 × ₹ 27.75 + ₹ 1000 + ₹ 1500)]

2400[₹ 26,000 - ₹ 24,700] = ₹ 31,20,000

LOSS DUE TO TRANSACTION EXPOSURE

₹ 34,80,000 – ₹ 31,20,000 = ₹ 3,60,000

(ii) Profit based on new exchange rates

2400[₹ 25,000 - (800 × ₹ 27.15 + ₹ 1,000 + ₹ 1,500)]

2400[₹ 25,000 - ₹ 24,220] = ₹ 18,72,000

Profit after change in exchange rates at the end of six months

2400 [₹ 25,000 - (800 × ₹ 27.75 + ₹ 1,000 + ₹ 1,500)]

2400 [₹ 25,000 - ₹ 24,700] = ₹ 7,20,000

Decline in profit due to transaction exposure

₹ 18,72,000 - ₹ 7,20,000 = ₹ 11,52,000

Current price of each unit in € = $\frac{₹\ 25,000}{₹\ 51.50}$ = € 485.44

Price after change in Exch. Rate = $\frac{₹\ 25,000}{₹\ 51.75}$ = € 483.09

Change in Price due to change in Exch. Rate

€ 485.44 - € 483.09 = € 2.35 or (-) 0.48%

Price elasticity of demand = 1.5

Increase in demand due to fall in price 0.48 × 1.5 = 0.72%

Size of increased order = 2400 ×1.0072 = 2417 units

Profit = 2417 [₹ 25,000 – (800 × ₹ 27.75 + ₹ 1,000 + ₹ 1,500)]

= 2417 [₹ 25,000 - ₹ 24,700] = ₹ 7,25,100

Therefore, decrease in profit due to operating exposure

₹ 18,72,000 – ₹ 7,25,100 = ₹ 11,46,900

Alternatively, if it is assumed that Fixed Cost shall not be changed with change in units then answer will be as follows:

Fixed Cost = 2400 [₹ 1,000] = ₹ 24,00,000

Profit = 2417 [₹ 25,000 – (800 × ₹ 27.75 + ₹ 1,500)] – ₹ 24,00,000

= 2417 (₹ 1,300) – ₹ 24,00,000 = ₹ 7,42,100

Therefore, decrease in profit due to operating exposure ₹ 18,72,000 – ₹ 7,42,100 = ₹ 11,29,900

Practical Q49

49. Your bank's London office has surplus funds to the extent of USD 5,00,000/- for a period of 3 months. The cost of the funds to the bank is 4% p.a. It proposes to invest these funds in London, New York or Frankfurt and obtain the best yield, without any exchange risk to the bank. The following rates of interest are available at the three centres for investment of domestic funds there at for a period of 3 months.

London	5 % p.a.
New York	8% p.a.
Frankfurt	3% p.a.

The market rates in London for US dollars and Euro are as under:

London on New York	
Spot	1.5350/90
1 month	15/18
2 months	30/35
3 months	80/85
London on Frankfurt	
Spot	1.8260/90
1 month	60/55
2 months	95/90
3 months	145/140

At which centre, will be investment be made & what will be the net gain (to the nearest pound) to the bank on the invested funds?

Solution — ICAI

49. (i) If investment is made at London

Convert US$ 5,00,000 at Spot Rate (5,00,000/1.5390)		= £ 3,24,886
Add: £ Interest for 3 months on £ 324,886 @ 5%		= £ 4,061
		= £ 3,28,947
Less: Amount Invested	$ 5,00,000	
Interest accrued thereon	$ 5,000	
	= $ 5,05,000	
Equivalent amount of £ required to pay the above sum ($ 5,05,000/1.5430*)		= £ 3,27,285
Arbitrage Profit		= £ 1,662

(ii) If investment is made at New York

Gain $ 5,00,000 (8% - 4%) x 3/12	= $ 5,000
Equivalent amount in £ 3 months ($ 5,000/ 1.5475)	£ 3,231

(iii) If investment is made at Frankfurt

Convert US$ 500,000 at Spot Rate (Cross Rate) 1.8260/1.5390	= € 1.1865
Euro equivalent US$ 500,000	= € 5,93,250
Add: Interest for 3 months @ 3%	= € 4,449
	= € 5,97,699
3 month Forward Rate of selling € (1/1.8150)	= £ 0.5510
Sell € in Forward Market € 5,97,699 x £ 0.5510	= £ 3,29,332
Less: Amounted invested and interest thereon	= £ 3,27,285
Arbitrage Profit	= £ 2,047

Since out of three options the maximum profit is in case investment is made in New York. Hence it should be opted.

* Due to conservative outlook.

Practical QSO

50. Drilldip Inc. a US based company has a won a contract in India for drilling oil field. The project will require an initial investment of ₹ 500 crore. The oil field along with equipments will be sold to Indian Government for ₹ 740 crore in one year time. Since the Indian Government will pay for the amount in Indian Rupee (₹) the company is worried about exposure due exchange rate volatility.

You are required to:

(a) Construct a swap that will help the Drilldip to reduce the exchange rate risk.

(b) Assuming that Indian Government offers a swap at spot rate which is 1US$ = ₹ 50 in one year, then should the company should opt for this option or should it just do nothing. The spot rate after one year is expected to be 1US$ = ₹ 54. Further you may also assume that the Drilldip can also take a US$ loan at 8% p.a.

Solution — ICAI

50. (a) The following swap arrangement can be entered by Drilldip.

(i) Swap a US$ loan today at an agreed rate with any party to obtain Indian Rupees (₹) to make initial investment.

(ii) After one year swap back the Indian Rupees with US$ at the agreed rate. In such case the company is exposed only on the profit earned from the project.

(b) With the swap

	Year 0 (Million US$)	Year 1 (Million US$)
Buy ₹ 500 crore at spot rate of 1US$ = ₹ 50	(100.00)	----
Swap ₹ 500 crore back at agreed rate of ₹ 50	----	100.00
Sell ₹ 240 crore at 1US$ = ₹ 54	----	44.44
Interest on US$ loan @8% for one year	----	(8.00)
	(100.00)	136.44

Net result is a net receipt of US$ 36.44 million.

Without the swap

	Year 0 (Million US$)	Year 1(Million US$)
Buy ₹ 500 crore at spot rate of 1US$ = ₹ 50	(100.00)	----
Sell ₹ 740 crore at 1US$ = ₹ 54	----	137.04
Interest on US$ loan @8% for one year	----	(8.00)
	(100.00)	129.04

Net result is a net receipt of US$ 29.04 million.

Decision: Since the net receipt is higher in swap option the company should opt for the same.

Practical QSI

51. You as a dealer in foreign exchange have the following position in Swiss Francs on 31st October, 2009:

	Swiss Francs
Balance in the Nostro A/c Credit	1,00,000
Opening Position Overbought	50,000
Purchased a bill on Zurich	80,000
Sold forward TT	60,000
Forward purchase contract cancelled	30,000
Remitted by TT	75,000
Draft on Zurich cancelled	30,000

What steps would you take, if you are required to maintain a credit Balance of Swiss Francs 30,000 in the Nostro A/c and keep as overbought position on Swiss Francs 10,000?

Solution — ICAI

51. Exchange Position:

Particulars	*Purchase Sw. Fcs.*	*Sale Sw. Fcs.*
Opening Balance Overbought	50,000	
Bill on Zurich	80,000	
Forward Sales – TT		60,000
Cancellation of Forward Contract		30,000
TT Sales		75,000
Draft on Zurich cancelled	30,000	—
	1,60,000	1,65,000
Closing Balance Oversold	5,000	—
	1,65,000	1,65,000

Cash Position (Nostro A/c)

	Credit	Debit
Opening balance credit	1,00,000	—
TT sales	—	75,000
	1,00,000	75,000
Closing balance (credit)	—	25,000
	1,00,000	1,00,000

The Bank has to buy spot TT Sw. Fcs. 5,000 to increase the balance in Nostro account to Sw. Fcs. 30,000.

This would bring down the oversold position on Sw. Fcs. as Nil.

Since the bank requires an overbought position of Sw. Fcs. 10,000, it has to buy forward Sw. Fcs. 10,000.

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